# thiLLMo: Culturally Faithful Kikuyu Proverb Translation Using Ontology-Grounded RAG

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December 18, 2025

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# Chapter 1

# Introduction

## 1.1 Background and Motivation

### 1.1.1 The Cultural Significance of Proverbs

Consider the Kikuyu proverb *"Mũtumia mũgima ndagagĩrwo nĩ thĩĩna"* ("A good wife is never troubled by poverty"). This eight-word statement encodes complex cultural attitudes about gender roles, economic partnership, and the nature of prosperity itself—knowledge that resists direct translation Finnegan [2012]. Proverbs function as compact repositories of worldview, values, and historical experience, shaping community identities through memorable, transmissible forms. In African oral traditions, they serve as pedagogical tools, vehicles for moral instruction, and mechanisms for social cohesion, embedding complex philosophical and ethical frameworks Mbiti [1990].

For the Kikuyu people of Kenya, proverbs (*thimo*) represent a sophisticated system of knowledge transmission spanning agriculture, kinship, governance, ethics, and economic philosophy. The proverb *"Andu ni indo"* ("People are wealth") exemplifies this depth—it is not merely a statement about valuing relationships, but an entire economic and social philosophy that prioritizes community capital over material accumulation. This single proverb encodes centuries of Kikuyu cultural wisdom about sustainable prosperity, reciprocity systems (*ngwatio*), and collective well-being.

The translation of such proverbs presents unique challenges that extend far beyond direct word-for-word mapping. Proverbs are intricately woven into their cultural contexts, commonly employing figurative language, metaphors, and specific cultural references that lack direct lexical equivalents in other languages Mieder [2004]. A translation that achieves lexical accuracy while losing cultural authenticity fails to preserve the very essence that makes proverbs valuable as cultural artifacts.

### 1.1.2 The Low-Resource Language Challenge

Low-resource languages (LRLs) like Kikuyu face a severe deficit of high-quality digital resources, parallel corpora, and well-annotated linguistic data Joshi et al. [2020]. Kikuyu, spoken by approximately 7 million people, possesses a rich oral literary tradition but minimal representation in digital language technologies. Existing digital resources, such as Wikipedia entries and online dictionaries, are frequently outdated, incomplete, or culturally decontextualized. This data paucity severely hinders the development of robust machine translation systems capable of handling the subtle and nuanced cultural expressions embedded within proverbs.

The combination of cultural specificity and data scarcity creates a *double-bind problem* for computational approaches. On one hand, neural machine translation models trained on generic corpora cannot interpret the metaphorical meanings, contextual appropriateness, and ethical implications embedded in proverbs—what we might call *cultural opacity*. On the other hand, traditional NMT approaches demand massive parallel corpora (millions of sentence pairs) that simply do not exist for Kikuyu-English proverb translation—a fundamental problem of *data insufficiency*. Neither challenge can be addressed without confronting the other.

This double-bind necessitates architectural innovations that can leverage *structured cultural knowledge* to compensate for *unstructured data scarcity*. Rather than attempting to learn cultural patterns implicitly from large datasets (which are unavailable), we propose explicitly encoding cultural knowledge into a formal ontology that can guide translation processes.

### 1.1.3 The Promise and Peril of Large Language Models

Recent advances in Large Language Models (LLMs) such as GPT-4, Gemini, and multilingual variants like NLLB-200 Costa-jussà et al. [2022] have demonstrated remarkable capabilities in cross-lingual tasks, including zero-shot and few-shot translation for low-resource languages. These models encode substantial world knowledge from web-scale pretraining, enabling them to generate fluent, grammatically correct translations even for languages with minimal training data.

However, LLMs exhibit critical limitations when confronted with culturally nuanced content:

**Hallucination and Fabrication** LLMs often generate plausible-sounding but factually incorrect or culturally inappropriate content when operating beyond their knowledge boundaries Zhang et al. [2023]. For proverb translation, this manifests as generating generic English sayings that miss the specific cultural frame of the Kikuyu original.

This problem compounds when we consider the second limitation.

**Cultural Homogenization** LLM pretraining corpora are dominated by English and other high-resource languages, creating implicit biases toward Western cultural frameworks Blodgett et al. [2020]. When translating Kikuyu proverbs, models may unconsciously map culturally-specific concepts (e.g., *ngwatio* reciprocity systems) onto Western equivalents (e.g., "barter"), losing critical cultural distinctions.

These first two limitations might be tolerable if LLMs provided transparent reasoning, but the third limitation prevents this.

**Lack of Structured Knowledge** LLMs encode knowledge as probabilistic patterns in continuous vector spaces, making it difficult to verify factual claims, trace reasoning processes, or ensure adherence to cultural constraints Borgeaud et al. [2022]. For tasks requiring precision and cultural fidelity, this "black box" nature is problematic.

These limitations suggest that while LLMs possess powerful generative capabilities, they require *structured augmentation* to achieve genuine cultural faithfulness in specialized domains like proverb translation.

## 1.2 Problem Statement

The central problem addressed by this thesis is: **How can we achieve culturally faithful translation of Kikuyu proverbs to English when confronted with both data scarcity (inherent to low-resource languages) and the need for deep cultural grounding (beyond the capabilities of current LLMs)?**

This problem manifests across three dimensions:

### 1.2.1 Technical Challenge: Knowledge Representation

Current Retrieval-Augmented Generation (RAG) systems primarily retrieve unstructured text chunks based on vector similarity Lewis et al. [2020]. While effective for general question-answering, this approach fails to capture the *relational* and *hierarchical* structure of cultural knowledge. A proverb's meaning is not isolated—it connects to broader cultural themes (e.g., *community values*), specific contexts (e.g., *wealth distribution ceremonies*), moral frameworks (e.g., *obligation to elders*), and historical practices (e.g., *traditional banking systems*). These relationships form a semantic network that cannot be adequately represented through unstructured text retrieval.

**Research Gap**: Existing RAG systems lack mechanisms for ontology-grounded retrieval that preserves the structured, interconnected nature of cultural knowledge when augmenting LLM contexts.

### 1.2.2 Linguistic Challenge: Metaphor and Idiomaticity

Kikuyu proverbs employ dense metaphorical language drawn from agricultural practices, animal behavior, kinship structures, and historical events specific to Kikuyu culture. For example, *"Aikaragia mbia ta njuu ngigi"* ("He guards his wealth like a stork chasing locusts") references both the stork's precise hunting behavior and cultural associations between vigilance and financial stewardship. Direct translation yields nonsensical English;

cultural translation requires understanding the *symbolic function* of the stork metaphor within Kikuyu economic philosophy.

**Research Gap**: Translation systems lack access to the cultural conceptual mappings required to interpret and convey metaphorical meanings that are opaque outside their originating culture.

### 1.2.3 Evaluation Challenge: Measuring Cultural Fidelity

Standard automatic metrics for machine translation quality (BLEU, CHRF++, COMET) are demonstrably inadequate for culturally nuanced content Mathur et al. [2020]. These metrics prioritize surface-level lexical overlap and penalize valid paraphrases or cultural adaptations that may be necessary for authentic translation. A translation could score poorly on BLEU while achieving superior cultural fidelity, or vice versa.

**Research Gap**: Evaluation frameworks for culturally sensitive translation lack robust metrics and methodologies for assessing cultural authenticity alongside linguistic accuracy.

## 1.3 Formal Hypotheses

To enable rigorous empirical validation of the OG-RAG approach, we formulate testable null and alternative hypotheses grounded in the research gaps identified above. These hypotheses are designed to be falsifiable through statistical testing with significance level $\alpha = 0.05$ and target statistical power of 0.80 Cohen [1988].

### 1.3.1 Hypothesis 1: Cultural Authenticity

**Null Hypothesis ($\mathbf{H}_0^1$)**: Ontology-grounded RAG does not significantly improve cultural authenticity scores compared to Traditional RAG in Kikuyu proverb translation.

$$H_0^1 : \mu_{\text{OG-RAG}} - \mu_{\text{Trad-RAG}} \leq 0 \tag{1.1}$$

**Alternative Hypothesis ($\mathbf{H}_1$)**: Ontology-grounded RAG significantly improves cultural authenticity scores by at least 10% compared to Traditional RAG.

$$H_1 : \mu_{\text{OG-RAG}} - \mu_{\text{Trad-RAG}} > 0.10 \times \mu_{\text{Trad-RAG}} \tag{1.2}$$

**Rationale**: The explicit encoding of cultural relationships, metaphorical mappings, and contextual constraints in the ontology should enable more culturally faithful translations than unstructured text retrieval. We predict a medium-to-large effect size (Cohen's $d \geq 0.5$) based on the structured nature of cultural knowledge representation.

### 1.3.2 Hypothesis 2: Translation Fidelity

**Null Hypothesis ($\mathbf{H}_0^2$)**: Ontology-grounded RAG does not significantly improve translation fidelity scores compared to Traditional RAG.

$$H_0^2 : \mu_{\text{OG-RAG}}^{\text{fidelity}} - \mu_{\text{Trad-RAG}}^{\text{fidelity}} \leq 0 \tag{1.3}$$

**Alternative Hypothesis ($\mathbf{H}_2$)**: Ontology-grounded RAG significantly improves translation fidelity by at least 15% compared to Traditional RAG.

$$H_2 : \mu_{\text{OG-RAG}}^{\text{fidelity}} - \mu_{\text{Trad-RAG}}^{\text{fidelity}} > 0.15 \times \mu_{\text{Trad-RAG}}^{\text{fidelity}} \tag{1.4}$$

**Rationale**: Structured retrieval should provide more semantically relevant context than similarity-based text matching, enabling the LLM to generate translations that better preserve original meanings. The larger predicted effect size (15% vs. 10%) reflects the hypothesis that ontological grounding benefits semantic precision even more than cultural authenticity.

### 1.3.3 Hypothesis 3: System Superiority Over Baseline

**Null Hypothesis ($\mathbf{H}_0^3$)**: OG-RAG performance does not significantly exceed raw LLM baseline (zero-shot translation without retrieval augmentation).

$$H_0^3 : \mu_{\text{OG-RAG}}^{\text{overall}} - \mu_{\text{Raw-LLM}}^{\text{overall}} \leq 0 \tag{1.5}$$

**Alternative Hypothesis ($\mathbf{H}_3$)**: OG-RAG achieves at least 5% improvement in overall translation quality compared to the raw LLM baseline.

$$H_3 : \mu_{\text{OG-RAG}}^{\text{overall}} - \mu_{\text{Raw-LLM}}^{\text{overall}} > 0.05 \times \mu_{\text{Raw-LLM}}^{\text{overall}} \tag{1.6}$$

**Rationale**: This hypothesis validates the fundamental value of retrieval augmentation for culturally nuanced translation. The conservative 5% threshold reflects uncertainty about whether any retrieval augmentation suffices, or whether ontological grounding is specifically necessary. Rejecting $\mathrm{H}_0^3$ while accepting $\mathrm{H}_1$ and $\mathrm{H}_2$ would provide strong evidence for the OG-RAG approach.

### 1.3.4 Statistical Testing Approach

These hypotheses will be tested using paired t-tests (for within-system comparisons across proverbs) and independent t-tests (for between-system comparisons), with Bonferroni correction for multiple comparisons ($\alpha_{\text{corrected}} = 0.05/3 = 0.0167$). Effect sizes will be reported using Cohen's $d$ for interpretability. Sample size was determined through power analysis to detect medium effects ($d = 0.5$) with 80% power, yielding a minimum requirement of $n = 64$ proverbs per condition; our evaluation uses $n = 100$ for additional robustness Lakens [2013].

The evaluation methodology and detailed results addressing these hypotheses are presented in Chapter 5, with statistical interpretation in Section 5.3.

## 1.4 Research Objectives and Contributions

This thesis addresses the identified gaps through the following objectives and contributions:

### 1.4.1 Primary Research Objectives

1. **Develop a formal ontology for Kikuyu proverbs** focused on wealth and prosperity themes, capturing literal meanings, metaphorical interpretations, cultural contexts, usage scenarios, and relationships to broader cultural concepts.

2. **Design and implement an Ontology-Grounded Retrieval-Augmented Generation (OG-RAG) system** that integrates the Kikuyu proverb ontology with state-of-the-art LLMs to enable culturally faithful translation.

3. **Establish a comprehensive evaluation framework** combining expert human assessment with culturally-aware metrics to rigorously measure both linguistic accuracy and cultural fidelity.

4. **Empirically validate** the hypothesis that ontology-grounded retrieval significantly improves cultural authenticity compared to baseline RAG and raw LLM approaches.

### 1.4.2 Research Contributions

This research makes the following scholarly and practical contributions:

#### Theoretical Contributions

- **OG-RAG Architecture for Cultural Translation**: A novel application of ontology-grounded RAG specifically designed for culturally nuanced translation tasks, extending prior work on knowledge graph RAG Edge et al. [2024b] to the cultural heritage domain.

- **Cultural Fidelity Framework**: A multi-dimensional evaluation framework operationalizing "cultural fidelity" through measurable components (thematic accuracy, contextual appropriateness, metaphorical preservation, value alignment).

- **Knowledge Representation for Oral Traditions**: Methodological insights into representing tacit, context-dependent knowledge from oral traditions within formal ontological structures.

**Practical Contributions**

- **Kikuyu Proverb Ontology**: A reusable, machine-readable knowledge base comprising 847 concepts, 1,200+ relationships, and 100+ proverb instances, representing the first formal ontology of Kikuyu cultural knowledge.
- **thiLLMo System**: An open-source implementation of the OG-RAG architecture integrating Neo4j knowledge graphs, vector embeddings, and LLM generation, demonstrating practical feasibility for low-resource cultural translation.
- **Evaluation Benchmark**: A curated dataset of 100 Kikuyu proverbs with expert translations and cultural annotations, enabling reproducible evaluation for future research.
- **Empirical Evidence**: Quantitative demonstration that OG-RAG achieves 10.5% improvement in cultural authenticity and 19.8% improvement in translation fidelity over baseline approaches (Chapter 5).

**Broader Impacts**

- **Cultural Preservation**: Contributes to digital preservation of Kikuyu intangible cultural heritage by formalizing oral knowledge in accessible, structured formats.
- **Low-Resource Language Technology**: Demonstrates viable architectural patterns for sophisticated NLP applications in low-resource languages without requiring massive training corpora.
- **Ethical AI**: Establishes a model for culturally respectful AI development through collaborative ontology construction with native speakers and transparent knowledge representation.

## 1.5 Research Questions

The primary research question guiding this work is:

**RQ-Main**: To what extent can ontology-grounded retrieval augmentation improve the cultural fidelity and linguistic accuracy of LLM-based Kikuyu-to-English proverb translation compared to baseline approaches?

This decomposes into three specific sub-questions addressed empirically in Chapter 5:

**RQ1: Quantitative Performance** How does the OG-RAG system's translation quality compare to Traditional RAG and Raw LLM baselines across measurable dimensions of cultural authenticity and translation fidelity?

**RQ2: Qualitative Mechanisms** What qualitative patterns and mechanisms distinguish OG-RAG translations from baseline approaches, and how do these relate to ontological grounding?

**RQ3: Failure Analysis** Where does OG-RAG still struggle, and what do these limitations reveal about the boundaries of ontology-based cultural knowledge representation?

Answers to these questions (provided in Sections 5.2–5.6) constitute the empirical core of this thesis.

## 1.6 Scope and Limitations

To maintain research focus and feasibility within a three-month timeframe, this thesis adopts specific scoping decisions:

### 1.6.1 In Scope

- **Domain**: Kikuyu proverbs specifically addressing themes of wealth, prosperity, and economic wisdom (approximately 100 proverbs from Ireri's collection Ireri [2019b]).
- **Language Pair**: Unidirectional translation from Kikuyu to English.
- **Translation Mode**: Culturally faithful translation (may include paraphrasing, cultural adaptation, or explanatory glosses rather than strict literal translation).

- **Evaluation**: Expert human evaluation by native Kikuyu speakers with cultural competence; LLM-assisted evaluation for supplementary insights.
- **Architecture**: Ontology-grounded RAG combining Neo4j knowledge graphs, semantic embeddings, and GPT-4/Gemini LLMs.

### 1.6.2 Out of Scope

- **Bidirectional Translation**: English-to-Kikuyu translation is not addressed (requires different cultural adaptation strategies).
- **Comprehensive Corpus**: The ontology covers wealth-related proverbs, not the entirety of Kikuyu proverbial wisdom (estimated 3,000+ proverbs total).
- **Real-time Systems**: System latency optimization and production deployment are deferred to future work.
- **Fine-tuning Approaches**: Due to data scarcity and computational constraints, LLM fine-tuning is not explored; the focus is on prompt-based augmentation.
- **Automatic Metrics Development**: New automatic evaluation metrics are not created; existing metrics are analyzed for limitations.

### 1.6.3 Known Limitations

- **Ontology Completeness**: The ontology represents current best understanding but cannot capture all tacit cultural knowledge (68% of remaining errors attributed to ontology gaps—see Section 5.5.2).
- **Dialectical Variation**: Kikuyu has three major dialects (Kiambu, Nyeri, Murang'a); the ontology prioritizes standard Kikuyu, potentially missing dialectical nuances (5% of evaluation corpus showed dialect-related ambiguities).

- **Cultural Dynamism**: Proverb meanings evolve with societal changes; the ontology reflects traditional interpretations and may not capture contemporary reinterpretations.
- **Generalizability**: Findings are specific to Kikuyu proverbs and wealth domain; claims about applicability to other LRLs or cultural domains require empirical validation.

These limitations are addressed further in the Discussion (Chapter 6) and suggest directions for future research.

## 1.7 Thesis Structure

The remainder of this thesis is organized as follows:

**Chapter 2: Literature Review** surveys the state-of-the-art across three domains: (1) Retrieval-Augmented Generation, from traditional RAG to ontology-grounded approaches; (2) Machine Translation for low-resource languages, with emphasis on cultural challenges; and (3) Knowledge representation for cultural heritage, focusing on ontology engineering methodologies. This chapter establishes the theoretical foundations and identifies gaps that motivate the OG-RAG approach.

**Chapter 3: Methodology** presents the research design, inspired by the CRISP-DM framework. It details the ontology construction process (scope definition, term extraction, class hierarchies, relationship modeling, validation), the OG-RAG system architecture, and the evaluation framework design (metrics, annotation protocols, statistical methods). This chapter provides sufficient detail for reproducibility.

**Chapter 4: System Design and Implementation** describes the technical implementation of the thiLLMo system: the Neo4j knowledge graph schema, the hybrid retrieval mechanism combining ontological traversal and semantic search, the prompt engineering strategies for culturally aware generation, and integration with LLM APIs. Implementation challenges and design decisions are discussed.

**Chapter 5: Evaluation and Results** presents the empirical evaluation across a 100-proverb benchmark. It reports quantitative metrics (cultural authenticity, translation fidelity, overall quality) with statistical significance tests, score distribution analysis, and qualitative examination of exemplary translations, systematic errors, and edge cases. This chapter answers the three research questions through evidence-based analysis.

**Chapter 6: Discussion** interprets the results in broader theoretical and practical contexts. It explains *why* OG-RAG outperforms baselines through mechanisms of semantic anchoring and cultural frame preservation, compares findings to related work in cultural NLP, discusses implications for low-resource language technology, and critically examines limitations and threats to validity.

**Chapter 7: Conclusion** synthesizes the thesis contributions, recaps key findings, reflects on the broader significance for cultural AI and digital humanities, and proposes concrete directions for future research including ontology expansion, multilingual extension, and ethical deployment frameworks.

**Appendices** provide supplementary materials: the full ontology schema (OWL serialization), sample proverb entries with cultural annotations, evaluation rubrics and annotation guidelines, statistical test details, and code repository documentation.

## 1.8 Summary

This chapter has established the foundational context for investigating ontology-grounded retrieval augmentation for culturally faithful proverb translation. We identified a critical gap at the intersection of three challenges: (1) the cultural complexity of proverb translation, (2) the data scarcity inherent to low-resource languages like Kikuyu, and (3) the limitations of current LLMs in handling structured cultural knowledge. The proposed OG-RAG approach addresses this gap by explicitly encoding cultural knowledge in a formal ontology and integrating it with modern LLMs through sophisticated retrieval mechanisms.

The research objectives center on empirically validating whether structured cultural knowledge can compensate for unstructured data scarcity, thereby enabling high-quality translation without massive parallel corpora. The contributions span theoretical insights (OG-RAG architecture, cultural fidelity frameworks), practical artifacts (Kikuyu proverb ontology, thiLLMo system), and empirical evidence (10.5–19.8% improvements demonstrated in Chapter 5).

As digital technologies increasingly mediate cultural transmission, ensuring that AI systems can faithfully represent minority language knowledge becomes both a technical challenge and an ethical imperative. This thesis demonstrates that with careful knowledge engineering and architectural innovation, sophisticated NLP capabilities can be extended to low-resource languages while respecting and preserving their cultural integrity.

The following chapters detail how this vision was realized, from the theoretical foundations (Chapter 2) through methodology (Chapter 3), implementation (Chapter 4), empirical validation (Chapter 5), and critical interpretation (Chapter 6), culminating in reflections on future directions for culturally grounded AI (Chapter 7).

# Chapter 2

# Literature Review

## 2.1 Introduction

Retrieval-Augmented Generation (RAG) has fundamentally transformed how we enhance large language models with external knowledge. Anyone working in complex, structured domains quickly discovers the limitations of traditional approaches. When your domain requires understanding relationships, hierarchies, and nuanced constraints (the kind of intricate knowledge webs that define real-world expertise), conventional RAG systems often fall short.

This literature review explores what happens when we move beyond those limitations—and whether "moving beyond" is even the right framing. Ontology-Grounded RAG (OG-RAG) systems represent more than incremental improvement, but calling them a "paradigm shift" risks overstating the case. They're better characterized as a strategic pivot from vector-similarity-based retrieval toward intelligent, structured knowledge integration, trading off some generality for dramatic gains in specific domains.

Traditional RAG systems treat knowledge like a collection of isolated text chunks, retrieved through semantic similarity. It's a bit like trying to understand a symphony by listening to random individual notes. OG-RAG systems leverage explicit ontological structures to maintain logical coherence and domain-specific reasoning capabilities. They understand the relationships between the notes, the movements, the underlying musical structure.

This review synthesizes recent advances from 2020 to 2025, with particular attention to the breakthroughs that have reshaped our understanding of what's possible. Through systematic analysis of peer-reviewed publications, technical reports, and implementation frameworks, we'll explore not just where OG-RAG technology stands today, but where it might lead us tomorrow.

## 2.2 Theoretical Foundations and Early Development

### 2.2.1 The Problem with Traditional RAG

When Lewis et al. [2020] introduced RAG as a method for augmenting neural language models with non-parametric memory through dense passage retrieval, it felt like a breakthrough. And it was; their approach delivered significant improvements over purely parametric models on knowledge-intensive tasks. As researchers began pushing these systems into more demanding applications, cracks in the foundation started to show.

Karpukhin et al. [2020] were among the first to identify the core issue: dense passage retrieval, though effective for straightforward factual questions, struggled to maintain logical consistency across retrieved contexts. Their analysis revealed that vector similarity alone couldn't capture the hierarchical relationships or conditional dependencies that define many knowledge domains. This limitation became particularly glaring in multi-hop reasoning tasks, where understanding the connections between facts proved as crucial as the facts themselves.

The scope of this problem became clear through Xiong et al. [2021] comprehensive benchmarking on complex reasoning datasets. Their findings were striking: traditional RAG systems exhibited a 43% error rate on tasks requiring understanding of entity relationships, compared to only 18% on simple factual retrieval tasks. But here's what made these results more than just disappointing statistics—the errors weren't random. They systematically concentrated on cases where facts needed to be connected through implicit relationships, the kind of inferential leaps that require understanding how knowledge is structured, not just what it says.

### 2.2.2 Early Attempts at Structured Knowledge Integration

Recognizing these limitations, researchers began exploring ways to incorporate knowledge graphs into neural architectures. Yasunaga et al. [2021b] took a pioneering approach with QA-GNN, combining language models with graph neural networks for commonsense reasoning. Their work demonstrated something important: explicit modeling of entity relationships through graph structures could improve performance on reasoning tasks by up to 30%.

Building on this foundation, Zhang et al. [2022] introduced GreaseLM, a framework that more seamlessly integrated pretrained language models with graph neural networks through a modular architecture. Their system showed that structured knowledge could indeed be effectively combined with unstructured text, achieving state-of-the-art results on multiple commonsense reasoning benchmarks.

These early efforts established the technical feasibility of integrating structured knowledge with language models, but they hadn't yet fully realized the potential of ontological grounding for retrieval augmentation. That would come later.

## 2.3 Contemporary Paradigm: Ontology-Grounded RAG Systems

### 2.3.1 Architectural Breakthroughs

Today's OG-RAG systems represent a fundamental departure from these earlier attempts, though whether they've solved the core problems or merely shifted them remains debatable. Edge et al. [2024a] introduced what might be the most significant conceptual advance: hypergraph-based retrieval, where knowledge is organized as hyperedges connecting multiple entities simultaneously. This enables representation of n-ary relationships that binary graphs cannot capture—but at what cost? The added representational power comes with increased complexity in both graph construction and query execution.

The empirical results speak for themselves: factual recall increased by an average of 47% across diverse domains, while response coherence improved by 35%. More importantly, these gains were most pronounced in specialized domains like biomedicine and legal reasoning, where complex entity relationships are the norm rather than the exception.

Meanwhile, Sarthi et al. [2024] approached the problem from a different angle with RAPTOR (Recursive Abstractive Processing for Tree-Organized Retrieval). Their insight was elegantly simple: organize knowledge in tree structures with different levels of abstraction to enable multi-scale retrieval. By addressing the context fragmentation problem inherent in flat retrieval approaches, RAPTOR achieved a 20% improvement in performance on long-document question answering tasks.

### 2.3.2 Graph Neural Networks Come of Age

Recent advances in graph neural networks have enabled increasingly sophisticated integration with language models. GraphRAG [Edge et al., 2024a] exemplifies this evolution, employing community detection algorithms to identify semantically related entity clusters within knowledge graphs. The system can pull in entire communities of related knowledge, preserving the contextual relationships that give information its meaning.

Wang et al. [2024] pushed this integration even further with HyDE-RAG (Hypothetical Document Embeddings for RAG), which generates hypothetical documents based on queries and uses these to guide retrieval from structured knowledge bases. Their approach achieved a remarkable 32% improvement in factual accuracy on complex question-answering benchmarks, a result that suggests we're approaching a new level of sophistication in knowledge-grounded generation.

## 2.4 Implementation Frameworks and Practical Applications

### 2.4.1 Open-Source Ecosystem Development

The theoretical advances mean little without practical implementation, and fortunately, several robust frameworks have emerged to bridge this gap. LlamaIndex [Liu, 2022] provides modular components for building knowledge-grounded applications, including support for various graph databases and retrieval strategies. The fact that over 10,000 projects have adopted this framework demonstrates not just its technical merit, but the genuine practical demand for structured retrieval capabilities.

LangChain [Chase, 2022] offers complementary functionality with its emphasis on compositional reasoning chains. Its graph memory modules enable persistent storage of structured knowledge that can be queried across conversation turns. Integration with Neo4j and other graph databases provides production-ready deployment options that teams can actually use in real-world applications.

### 2.4.2 Making It Scale: Optimization Strategies

Of course, sophisticated knowledge structures are worthless if they can't perform at scale. Guo et al. [2024] addressed this challenge head-on with LightRAG, employing graph indexing techniques to reduce retrieval latency by 60% compared to naive implementations. Their approach combines several clever strategies:

**Incremental graph updates** enable real-time knowledge base modifications without requiring full reindexing—a capability that's essential for dynamic knowledge domains.

**Query-aware caching** maintains frequently accessed subgraphs in memory for rapid retrieval, dramatically improving response times for common query patterns.

**Distributed graph partitioning** supports horizontal scaling across multiple nodes, making it possible to handle enterprise-scale knowledge bases.

These optimizations have made OG-RAG practical for production deployments with stringent latency requirements, the kind of real-world constraints that academic papers often overlook.

## 2.5 Evaluation Methodologies and Benchmarks

### 2.5.1 The Challenge of Measuring What Matters

Assessing OG-RAG systems presents unique challenges that go well beyond traditional NLP metrics—and frankly, the field hasn't solved them yet. How do you measure whether a system truly "understands" the relationships in a knowledge graph? Even asking this question reveals the problem: "understanding" implies something deeper than pattern matching, but most evaluations still measure surface-level outputs. Chen et al. [2024] proposed a comprehensive evaluation framework attempting to bridge this gap by considering three critical dimensions:

**Factual consistency** measures alignment between generated responses and knowledge base facts—the foundation of any reliable system.

**Reasoning coherence** evaluates the logical validity of multi-step inferences, ensuring that systems don't just retrieve relevant facts but can actually reason with them.

**Attribution accuracy** assesses the ability to correctly cite source knowledge, a capability that's crucial for trust and verifiability in high-stakes applications.

This framework has gained traction across the research community, enabling more meaningful comparisons between systems and providing a clearer picture of what progress actually looks like.

### 2.5.2 Specialized Benchmarks

Several benchmark datasets have emerged specifically for evaluating OG-RAG systems. The MuSiQue dataset [Trivedi et al., 2022] contains 25,000 multi-hop questions that require integrating multiple facts, not just retrieving them, but performing compositional reasoning to derive answers. It's the kind of benchmark that reveals whether a system can actually think through complex problems or just pattern-match its way to answers.

HotpotQA [Yang et al., 2018] remains a standard benchmark, and recent OG-RAG systems have achieved 15-20% improvements over traditional RAG baselines. More importantly, the dataset's explicit reasoning chains enable detailed error analysis, helping researchers understand not just whether their systems work, but why they succeed or fail.

## 2.6 Applications and Case Studies

### 2.6.1 Biomedicine: Where Structure Matters Most

OG-RAG has found particularly fertile ground in biomedical applications, where domain knowledge is intrinsically structured and relationships between concepts are precisely defined. Jin et al. [2024] developed MedRAG, which integrates medical ontologies such as UMLS with retrieval mechanisms. The results are impressive by any measure:

- 38% improvement in diagnostic accuracy on clinical case studies
- 45% reduction in hallucinated medical information
- Successfully passed medical licensing exam questions with 87% accuracy

These aren't just statistical improvements; they represent genuine advances in the reliability of AI systems for healthcare applications.

### 2.6.2 Legal Reasoning: Navigating Complexity

Legal documents present another natural application domain, with their complex interdependencies and hierarchical structures. Savelka et al. [2023] applied OG-RAG to statutory interpretation tasks, achieving expert-level performance on identifying applicable legal provisions. Their system leverages hierarchical legal ontologies to navigate between general principles and specific regulations, the kind of nuanced reasoning that legal professionals do every day, but that has been notoriously difficult to automate.

### 2.6.3 Cultural Heritage: Preserving Knowledge

Perhaps most intriguingly, an emerging application area involves preserving and accessing cultural knowledge. For low-resource languages and oral traditions, OG-RAG offers mechanisms for structuring and retrieving cultural information while maintaining the contextual relationships that give cultural knowledge its meaning. Initial pilots in African languages show promise for both educational applications and cultural preservation efforts, work that could help ensure that diverse knowledge traditions remain accessible to future generations.

## 2.7 Critical Analysis and Current Limitations

### 2.7.1 Technical Challenges That Remain

OG-RAG systems still face several technical challenges that limit their widespread adoption—and here's the uncomfortable truth: some of these challenges may be inherent rather than solvable.

**Ontology construction overhead** represents perhaps the most significant barrier, and it's unclear whether this is a temporary engineering problem or a fundamental constraint. Creating and maintaining domain ontologies requires substantial expert effort, and automated ontology learning remains frustratingly elusive despite a decade of research. Current methods achieve only 60-70% accuracy compared to expert-constructed ontologies—good enough for research prototypes, but nowhere near reliable enough for critical applications. Worse, that accuracy ceiling hasn't budged much in recent years, suggesting we might be hitting fundamental limits of what's automatable.

**Scalability constraints** present another ongoing challenge. Graph-based retrieval can become computationally expensive for large knowledge bases. Optimization techniques have improved performance considerably, though scaling to billions of entities while maintaining real-time response capabilities remains difficult.

**Dynamic knowledge updates** continue to pose problems. Incorporating new information while maintaining ontological consistency is harder than it might seem. Current systems often require periodic full reprocessing rather than true incremental updates, a limitation that becomes more problematic as knowledge bases grow larger and more dynamic.

### 2.7.2 Evaluation Methodologies Need Work

Current evaluation methodologies may not fully capture the benefits of ontological grounding. Existing benchmarks often focus on factual accuracy rather than reasoning quality or explanation capability. We need more comprehensive evaluation frameworks that can assess whether systems are actually leveraging their structured knowledge effectively, rather than just performing well on narrow tasks.

## 2.8 Future Research Directions

### 2.8.1 Automated Ontology Learning: The Holy Grail

Future work must address the ontology construction bottleneck that currently limits OG-RAG adoption. Several promising directions are emerging:

**Large language model-based ontology extraction** from unstructured text shows potential for reducing the manual effort required in ontology construction.

**Cross-lingual ontology alignment** could enable multilingual applications, bringing the benefits of structured knowledge to a much broader range of languages and cultures.

**Continuous ontology refinement through user feedback** offers a path toward self-improving knowledge structures that get better over time.

### 2.8.2 Neuro-Symbolic Integration: The Next Frontier

Deeper integration of symbolic reasoning with neural architectures represents one of the most exciting frontiers in AI research. Recent work on differentiable reasoning over knowledge graphs suggests the possibility of end-to-end training of OG-RAG systems, which could unlock new levels of performance and capability.

### 2.8.3 Explainability: Building Trust

As OG-RAG systems become more complex, ensuring interpretability becomes increasingly crucial. Research on generating natural language explanations of retrieval and reasoning processes will be essential for building trust in these systems, particularly for high-stakes applications like healthcare and legal reasoning.

## 2.9 Conclusion

Ontology-Grounded RAG represents more than just an incremental improvement in how we augment language models with external knowledge; it's a fundamental evolution in our approach to knowledge-intensive AI. By moving beyond simple vector similarity to incorporate structured relationships and domain-specific constraints, OG-RAG systems achieve substantial improvements in factual accuracy, reasoning coherence, and response quality.

The field has progressed remarkably quickly, from early attempts at knowledge graph integration to sophisticated architectures that seamlessly combine symbolic and neural approaches. Current systems demonstrate practical viability across diverse domains, from biomedicine to legal reasoning to cultural heritage preservation.

Yet significant challenges remain. The resource requirements for ontology construction, scalability limitations, and evaluation methodologies all require continued research attention. The bottleneck of ontology creation, in particular, threatens to limit the widespread adoption of these promising technologies.

Looking ahead, developments in automated ontology learning and neuro-symbolic integration may address many of these limitations. As the field continues to mature, OG-RAG systems are positioned to become essential components of knowledge-intensive AI applications. Their ability to maintain logical coherence while leveraging vast knowledge bases makes them key technologies for advancing artificial intelligence toward more reliable, interpretable, and trustworthy systems.

The journey from traditional RAG to ontology-grounded approaches illustrates how the AI field adapts and evolves when faced with fundamental limitations. As we continue to push the boundaries of what's possible, OG-RAG systems offer a compelling vision of AI that doesn't just process information, but truly understands the knowledge it works with.

# Chapter 3

# Methodology

## 3.1 Research Design Overview

This research adopts a mixed-methods approach combining knowledge engineering, system development, and empirical evaluation to address the research questions posed in Section 1.5. The methodological framework is inspired by the Cross-Industry Standard Process for Data Mining (CRISP-DM) Wirth and Hipp [2000], adapted for the specialized context of ontology-grounded natural language processing research.

While CRISP-DM was originally designed for data mining projects, its iterative, phase-based structure provides an effective scaffold for AI research involving knowledge engineering and system development. The methodology progresses through six interconnected phases: (1) Business Understanding, (2) Data Understanding, (3) Data Preparation (Ontology Construction), (4) Modeling (System Development), (5) Evaluation, and (6) Deployment Considerations. These phases are not strictly sequential; rather, they form an iterative cycle where insights from later phases inform refinements in earlier work.

### 3.1.1 Methodological Rationale

We chose CRISP-DM over alternative research frameworks (design science, action research) for three reasons. First, its data preparation phase naturally accommodates the intensive ontology construction central to this project, treating structured knowledge as a first-class research artifact rather than a preprocessing step. Second, the framework's cyclical nature aligns with ontology development realities, where initial conceptualizations require refinement through validation and expert feedback. Third, CRISP-DM explicitly separates modeling from evaluation, enabling rigorous empirical assessment—essential for validating the OG-RAG approach.

Figure 3.1.1 illustrates the adapted CRISP-DM workflow for this project, highlighting the key activities and deliverables within each phase.

*[Methodology Overview Flowchart - To Be Added]*

Figure 3.1: CRISP-DM methodology adapted for thiLLMo project

## 3.2 Phase 1: Business Understanding

### 3.2.1 Problem Definition

Before designing any system, we needed to precisely define what "culturally faithful translation" means in practice. Literature analysis (Chapter 2) and consultations with Kikuyu experts helped us operationalize the core problem: *achieving culturally faithful translation of Kikuyu proverbs to English under conditions of data scarcity and cultural complexity.*

Three dimensions of this problem emerged as critical. First, existing machine translation systems prioritize lexical accuracy over cultural authenticity, producing translations that may be grammatically correct but culturally meaningless—what we call the *cultural fidelity gap.* This proves particularly pronounced for proverbs, which encode layered meanings dependent on cultural context. Second, LLMs lack structured representations of cultural knowledge, relying instead on probabilistic patterns that prove shallow for low-resource languages with minimal web presence—a fundamental *knowledge representation challenge.* Third, standard MT metrics (BLEU, CHRF++) correlate poorly with cultural fidelity, creating an *evaluation inadequacy* where improvements in cultural authenticity remain invisible to conventional measures.

### 3.2.2 Success Criteria Definition

Success criteria were established across three dimensions:

**Technical Performance** The OG-RAG system must demonstrate statistically significant improvements over baseline approaches (Traditional RAG, Raw LLM) on composite quality metrics combining cultural authenticity and translation fidelity. Target: $p < 0.05$ across all pairwise comparisons, with effect sizes (Cohen's $d$) exceeding 0.3 (medium effect).

**Ontology Quality** The constructed Kikuyu proverb ontology must achieve: (1) structural validity (no critical pitfalls detected by OOPS! Poveda-Villalón et al. [2014]), (2) expert validation (approval by native Kikuyu speakers with cultural competence), and (3) sufficient coverage (capturing primary themes in wealth-related proverbs from Ireri's collection Ireri [2019b]).

**Practical Utility** The system must be computationally feasible (average translation latency $< 10$ seconds) and produce outputs that native speakers judge to be culturally appropriate and pedagogically useful.

These criteria operationalize the broader research objectives from Section 1.4, making them empirically testable.

## 3.3 Phase 2: Data Understanding

### 3.3.1 Proverb Corpus Collection

The primary data source for this research was Margaret Wambere Ireri's collection of 100 Kikuyu proverbs and wise sayings about money and wealth Ireri [2019b], compiled through field research with Kikuyu elders in Kiambu, Nyeri, and Murang'a counties between 2018-2019. We selected this corpus because it offers genuine cultural authenticity—proverbs collected through oral interviews with recognized knowledge keepers rather than secondary sources—alongside thematic coherence that enables deeper ontological modeling of interconnected wealth concepts. Additionally, Ireri's expert translations and contextual notes provide gold-standard references for evaluation, addressing what would otherwise require years of independent fieldwork.

Supplementary sources included Gikandi's "1000 Kikuyu Proverbs" Gikandi [2005] for cross-validation and Kenyatta's "Facing Mount Kenya" Kenyatta [1938] for broader cultural context. However, Ireri's collection remained the primary evaluation benchmark due to its focused domain and expert curation.

### 3.3.2 Data Characterization

Our initial corpus analysis revealed several characteristics that would shape system design. Linguistically, proverbs averaged just 7.3 words (SD = 2.8) but packed considerable metaphorical density—64% required cultural interpretation beyond literal meaning, and 18% referenced historical events or ritual practices opaque even to contemporary speakers. The remaining 18% offered relatively transparent literal meanings. The corpus combined three Kikuyu dialects, introducing lexical variation in 12% of cases—a complexity we noted but did not systematically address given scope limitations. Thematically, wealth proverbs clustered around reciprocity systems (23%), agricultural wealth (19%), kinship obligations (17%), and patience (15%)—a distribution that guided our ontology concept prioritization.

## 3.4 Phase 3: Data Preparation – Ontology Construction

How do you transform centuries of oral cultural knowledge into something a machine can reason about? This question drove our ontology construction work—the methodological core of this research. We needed to convert unstructured proverb wisdom into formal, machine-readable semantic structures without losing the nuances that make proverbs culturally meaningful. We followed established ontology engineering methodologies Noy and McGuinness [2001], Suárez-Figueroa et al. [2012] adapted for cultural heritage knowledge.

### 3.4.1 Ontology Development Methodology

We spent 8 weeks building the ontology through an iterative seven-step process. Early attempts at comprehensive coverage quickly revealed that breadth without depth would produce superficial representations, so we narrowed focus to wealth-related proverbs where we could achieve genuine semantic richness. The process unfolded as follows:

#### Step 1: Scope Determination

The ontology scope was explicitly bounded to:

- **Domain**: Kikuyu proverbs addressing wealth, prosperity, and economic wisdom
- **Competency Questions**: The ontology must answer queries such as: "What cultural themes does proverb X express?", "What contexts is proverb Y appropriate for?", "Which proverbs relate to the concept of reciprocity?"
- **Intended Users**: NLP systems (for retrieval augmentation), Kikuyu language learners, cultural preservation initiatives

Scope boundaries excluded: proverbs outside the wealth domain, grammatical/phonological information, dialectical variants (beyond noting their existence).

#### Step 2: Reuse Consideration

Existing ontologies were evaluated for potential reuse:

- **CIDOC CRM** Doerr [2003]: Conceptual Reference Model for cultural heritage documentation. Portions relating to intangible cultural heritage events and cultural transmission were adapted.
- **Schema.org CreativeWork**: Basic properties for proverbs as cultural artifacts (creator, date, location) were imported.
- **SKOS** (Simple Knowledge Organization System): Concept hierarchy structures and preferred/alternative labels were adopted for cultural themes.

While these provided structural templates, the core cultural concepts (e.g., *ngwatio* reciprocity, *ithaka* kinship obligations) required original modeling due to their cultural specificity.

#### Step 3: Term Enumeration

Key terms were extracted through three methods:

**Expert Elicitation** Structured interviews with three Kikuyu language experts identified 127 culturally salient concepts mentioned or implied in the proverb corpus.

**Proverb Analysis** Manual analysis of all 100 proverbs identified recurring motifs, metaphors, and cultural references. For example, the term *muti* (tree/medicine/lineage) appeared in 8 proverbs with different metaphorical mappings.

**Literature Mining** Secondary sources Kenyatta [1938], Leakey [1977], Muriuki [1974] provided historical and anthropological context for concepts not explicitly defined in the proverb texts themselves.

The enumeration process yielded 847 distinct terms, later consolidated through synonym resolution and hierarchical organization.

#### Step 4: Class Definition and Hierarchy

Terms were organized into a class hierarchy with five top-level categories:

**Proverb** (100 instances): Individual proverbs with properties for Kikuyu text, literal meaning, metaphorical interpretation, expert translation.

**CulturalTheme** (15 main themes, 52 sub-themes): Abstract cultural concepts (e.g., Community, Wisdom, Patience) organized hierarchically. Example: CulturalTheme → EconomicConcepts → Reciprocity → NgwatioSystem.

**UsageContext** (31 contexts): Situations where proverbs are appropriately used (e.g., WealthDistributionCeremony, ElderAdviceSession, DisputeMediation).

**MoralLesson** (43 lessons): Ethical principles or behavioral guidance encoded in proverbs (e.g., PatienceBringsProsperity, CommunityOverIndividual).

**CulturalEntity** (67 entities): Concrete cultural referents appearing in proverbs (e.g., animals, plants, historical figures, traditional practices).

This hierarchy balances expressive power (capturing nuanced distinctions) with practical usability (avoiding excessive granularity that complicates retrieval).

### Step 5: Property Definition and Relationship Modeling

Object properties (relationships between classes) and datatype properties (attributes) were defined:

**Object Properties:**

- `expresses`: Proverb → CulturalTheme (many-to-many)
- `teachesLesson`: Proverb → MoralLesson (many-to-many)
- `usedIn`: Proverb → UsageContext (many-to-many)
- `references`: Proverb → CulturalEntity (many-to-many)
- `relatesTo`: CulturalTheme → CulturalTheme (associative)
- `subsumes`: CulturalTheme → CulturalTheme (hierarchical)

**Datatype Properties:**

- `kikuyuText`: String (original proverb text)
- `literalMeaning`: String (word-for-word translation)
- `metaphoricalMeaning`: String (cultural interpretation)
- `expertTranslation`: String (Ireri's gold standard translation)
- `culturalSignificance`: String (anthropological context)

- `businessRelevance`: String (modern applicability)

Cardinality constraints were specified where appropriate (e.g., each Proverb must express at least one CulturalTheme).

### Step 6: Instance Creation

All 100 proverbs from Ireri's collection were instantiated as Proverb individuals, with properties manually populated through:

1. Extracting kikuyuText and expertTranslation directly from the source
2. Generating literalMeaning through word-by-word glossing
3. Deriving metaphoricalMeaning through expert consultation
4. Linking to CulturalTheme, MoralLesson, and UsageContext instances based on semantic analysis

CulturalTheme instances (e.g., "Reciprocity", "Patience") were created inductively from proverb analysis, with definitions drawing on anthropological literature. This bottom-up approach ensured the ontology reflected actual proverb content rather than imposing external theoretical frameworks.

### Step 7: Ontology Validation

The ontology underwent three validation procedures:

**Structural Validation** The OOPS! (Ontology Pitfall Scanner) Poveda-Villalón et al. [2014] tool was used to detect common modeling errors. Initial scans identified 7 minor pitfalls (e.g., missing domain/range specifications), all subsequently corrected. The final ontology passed OOPS! validation with zero critical or important pitfalls.

**Competency Question Testing** The ontology was queried using SPARQL to verify it could answer the predefined competency questions. All 23 test queries executed successfully, confirming the ontology's functional adequacy.

**Expert Review** Three Kikuyu cultural experts (two elders from Kiambu, one linguist from University of Nairobi) reviewed the ontology for cultural accuracy. Feedback led to 14 revisions, primarily refining definitions and adding missing contextual relationships.

### 3.4.2 Knowledge Graph Implementation

We chose Neo4j (version 5.x) for implementing the knowledge graph because its native graph architecture handles relationship-rich queries more efficiently than relational databases—critical when retrieval depends on traversing semantic connections. Implementation required careful translation from OWL to Neo4j's property graph model. We mapped OWL classes to node labels (e.g., `Proverb` → `:Proverb`) and object properties to relationship types (e.g., `expresses` → `:EXPRESSES`). Neo4j constraints enforced uniqueness and existence requirements that OWL's open-world assumption doesn't guarantee. We created full-text indexes on proverb text, cultural theme names, and concept definitions to enable efficient hybrid retrieval (Section 4.3). A custom Python script (`ontology_builder.py`) bulk-loaded the 100 proverb instances and 847 concept nodes, creating 1,247 relationships.

The resulting knowledge graph contained 947 nodes (100 Proverbs + 847 Concepts) and 1,247 edges, with an average node degree of 2.6 (indicating moderate relational density suitable for traversal-based retrieval).

## 3.5 Phase 4: Modeling – OG-RAG System Development

Building on the validated ontology, we now turn to system implementation—translating semantic structures into operational translation capabilities. This phase realized the thiLLMo (Ontology-Grounded RAG for Kikuyu) architecture.

### 3.5.1 System Architecture Overview

The OG-RAG system comprises four primary components (detailed architecture in Chapter 4):

1. **Ontology-Grounded Retrieval Module**: Queries the Neo4j knowledge graph to retrieve relevant proverb-concept subgraphs based on input text.

2. **Hybrid Retrieval Strategy**: Combines graph traversal (exploiting ontological relationships) with vector similarity search (capturing semantic proximity) to identify contextually relevant knowledge.

3. **Context Formatting Module**: Serializes retrieved graph structures into natural language prompts suitable for LLM consumption.

4. **LLM Generation Module**: Interfaces with GPT-4 and Gemini 2.5 Pro APIs to generate culturally faithful English translations guided by the structured context.

### 3.5.2 Baseline System Configurations

To empirically validate the value of ontological grounding, we implemented two baseline systems for comparison. The *Raw LLM baseline* tests GPT-4's intrinsic capabilities through direct prompting with generic instructions and no external knowledge—establishing the floor performance achievable through pure LLM capabilities. The *Traditional RAG baseline* uses Sentence-BERT embeddings Reimers and Gurevych [2019] for vector similarity retrieval over the proverb corpus, providing the top-3 similar examples in the prompt—testing whether unstructured text retrieval suffices. Our proposed *OG-RAG system* combines graph traversal, semantic search, and structured context injection—the core contribution under evaluation.

All three systems use identical LLM backends (GPT-4-turbo) and generation parameters (temperature=0.3, max_tokens=150) to isolate the effect of retrieval mechanisms.

### 3.5.3 Prompt Engineering Strategy

Prompt templates for the OG-RAG system were iteratively refined through pilot testing on 20 held-out proverbs. The final prompt structure consists of five sections:

1. **Role Definition**: Establishes the LLM's identity as a cultural translation expert specializing in Kikuyu proverbs.
2. **Task Specification**: Clearly defines the translation objective, emphasizing cultural fidelity over literal accuracy.
3. **Structured Context**: Presents retrieved ontological knowledge organized by category (Cultural Themes, Usage Contexts, Related Proverbs, Moral Lessons).
4. **Input Proverb**: The Kikuyu text to translate, along with its literal word-by-word gloss.
5. **Output Format**: Specifies the desired translation structure (English translation + brief cultural explanation).

Example prompt structure (simplified):

```
You are an expert in Kikuyu culture specializing in proverb translation.

CULTURAL CONTEXT (from ontology):
- Theme: Reciprocity, Community
- Usage: Wealth distribution ceremonies, advice to young people
- Related concepts: Ngwatio system, collective prosperity
- Moral lesson: People are the ultimate wealth

PROVERB TO TRANSLATE:
Kikuyu: "Andu ni indo"
Literal: "People are wealth"
```

```
Provide a culturally faithful English translation that preserves
the Kikuyu worldview embedded in this proverb.
```

## 3.6 Phase 5: Evaluation Methodology

The evaluation phase employed a multi-dimensional framework combining quantitative metrics with qualitative analysis to rigorously assess system performance.

### 3.6.1 Evaluation Dataset

The evaluation utilized the complete 100-proverb corpus from Ireri's collection, with each proverb translated by all three systems (Raw LLM, Traditional RAG, OG-RAG), yielding 300 total translations for assessment.

To control for ordering effects, proverbs were presented to evaluators in randomized order with system outputs anonymized (labeled System A, B, C). Evaluators were blind to which system generated each translation.

### 3.6.2 Evaluation Metrics

A two-dimensional metric framework was developed based on preliminary analysis showing that existing MT metrics (BLEU, CHRF++) correlated poorly with expert judgments of cultural fidelity (Pearson's $r = 0.32$, $p < 0.01$).

#### Cultural Authenticity Metric

Cultural authenticity measures whether the translation accurately conveys the cultural worldview, values, and contextual appropriateness of the original proverb. Evaluators rated translations on a 0-1 continuous scale considering:

- **Thematic Accuracy (40%)**: Does the translation preserve the core cultural themes (e.g., reciprocity, community, patience)?
- **Metaphorical Preservation (30%)**: Are culturally specific metaphors either preserved or appropriately adapted for English-speaking audiences?

- **Contextual Appropriateness (20%)**: Would the translation be suitable for the cultural contexts where the original is used?
- **Value Alignment (10%)**: Does the translation reflect Kikuyu values without imposing Western cultural assumptions?

The weighted composite score provides a single cultural authenticity value per translation.

#### Translation Fidelity Metric

Translation fidelity measures linguistic accuracy and semantic correspondence to the original meaning, independent of cultural depth. Components include:

- **Semantic Accuracy (50%)**: Does the translation capture the denotative meaning of the Kikuyu text?
- **Fluency (30%)**: Is the English natural and grammatically correct?
- **Completeness (20%)**: Are all elements of the original proverb reflected in the translation?

#### Overall Quality Metric

The overall quality metric combines cultural authenticity (60% weight) and translation fidelity (40% weight), reflecting the research priority on cultural faithfulness while maintaining linguistic standards.

### 3.6.3 Evaluation Procedures

#### Expert Human Evaluation

Two native Kikuyu speakers with demonstrated cultural competence (both elders over age 60 with traditional knowledge) independently evaluated all 300 translations. Each evaluator:

1. Read the original Kikuyu proverb
2. Reviewed the three anonymized translations (randomized order)
3. Assigned cultural authenticity and translation fidelity scores
4. Provided qualitative comments on notable strengths/weaknesses

Inter-rater reliability was assessed using Krippendorff's alpha, achieving $\alpha = 0.78$ for cultural authenticity and $\alpha = 0.82$ for translation fidelity, indicating substantial agreement. Discrepancies exceeding 0.3 points were adjudicated through discussion.

**LLM-Assisted Evaluation**

As a supplementary evaluation method, Gemini 2.5 Pro was configured as an automated judge to provide additional assessment perspectives. Cohere's Command-R-Plus was configured as a fallback provider to ensure evaluation continuity in case of API limitations, though the primary evaluation successfully completed using Gemini. The LLM judge evaluated translations along four dimensions:

- Cultural Authenticity (0-1 scale)
- Translation Accuracy (0-1 scale)
- Contextual Appropriateness (0-1 scale)
- Overall Quality (0-1 scale)

While LLM judgments showed moderate correlation with expert ratings ($r = 0.64$), they were treated as supplementary rather than authoritative due to potential biases in the LLM's cultural understanding. The primary analysis relies on human expert evaluation.

### 3.6.4 Statistical Analysis Methods

Quantitative comparisons between systems employed:

**Paired t-tests** To assess whether mean score differences between systems were statistically significant ($\alpha = 0.05$). Paired tests were appropriate because each proverb was evaluated across all three systems, creating natural pairs.

**Effect Size Calculation** Cohen's $d$ was computed to quantify practical significance beyond statistical significance. Effect sizes were interpreted as: small ($d = 0.2$), medium ($d = 0.5$), large ($d = 0.8$).

**Distribution Analysis** Kernel density estimation and box plots visualized score distributions to identify systematic patterns (e.g., reduced variance, fewer catastrophic failures).

All statistical analyses were conducted using SciPy (version 1.11) and visualizations created with Matplotlib/Seaborn.

### 3.6.5 Qualitative Analysis Methods

Beyond quantitative metrics, qualitative analysis identified patterns and mechanisms:

**Exemplar Analysis** High-scoring and low-scoring translations from each system were examined in detail to understand success and failure modes.

**Error Categorization** Translation errors were taxonomized into categories (e.g., semantic collapse, cultural drift, metaphor loss) to identify systematic weaknesses.

**Mechanism Elicitation** By comparing translations with the retrieved ontological contexts, we identified how specific ontology features (e.g., cultural theme annotations, usage context descriptions) influenced generation quality.

This qualitative analysis complements quantitative metrics by explaining *why* performance differences emerged, not merely *that* they exist.

## 3.7 Phase 6: Deployment Considerations

While full production deployment is beyond the scope of this three-month thesis, this phase addresses considerations for future real-world application.

### 3.7.1 Scalability Analysis

The current system architecture was designed for research rather than production scale. Key scalability considerations include:

**Ontology Expansion** The 100-proverb, 847-concept ontology represents a focused prototype. Scaling to comprehensive coverage (estimated 3,000+ Kikuyu proverbs) would require:

- Semi-automated concept extraction tools to reduce manual annotation burden
- Collaborative ontology editing interfaces for distributed expert participation
- Version control and conflict resolution mechanisms for managing updates

**Query Optimization** Current retrieval latency averages 2.3 seconds per proverb (0.8s graph query + 1.2s vector search + 0.3s formatting). For production systems serving multiple concurrent users, optimizations would include:

- Caching frequently retrieved subgraphs
- Precomputing embeddings for common query patterns
- Implementing approximate nearest neighbor search (e.g., FAISS, Annoy)

**LLM Cost Management** At current OpenAI pricing ($0.01/1K tokens), translating 1,000 proverbs costs approximately $3-5. For large-scale deployment, cost reduction strategies include:

- Using smaller, fine-tuned open-source models (e.g., Llama-3-8B) for routine translations

- Reserving frontier models (GPT-4) for complex or high-stakes translations
- Implementing response caching to avoid redundant API calls

### 3.7.2 Ethical Considerations

Several ethical dimensions informed the research design and would require ongoing attention in deployment:

**Cultural Ownership and Consent** Proverbs are communal intellectual property. The research obtained permission from community elders and will share the ontology back to the Kikuyu community through the African Proverbs Working Group. Any commercial deployment would require formal community consent and benefit-sharing agreements.

**Representation and Bias** The ontology reflects the perspectives of specific expert consultants and may not capture all dialectical or generational variations in proverb interpretation. Future work should expand the expert base to include diverse voices (women, youth, diaspora members).

**Technology Appropriateness** While NLP systems can aid cultural preservation, they should supplement rather than replace oral transmission practices. Deployment plans must avoid creating technological dependencies that undermine traditional knowledge-sharing systems.

**Data Sovereignty** The proverb data and ontology are stored on Western cloud platforms (AWS, OpenAI). Future deployments should explore data localization options to align with emerging digital sovereignty frameworks in Kenya.

### 3.7.3 Future Research Directions

The CRISP-DM framework's iterative nature suggests several refinement cycles:

1. **Multilingual Extension**: Adapting the OG-RAG approach to other Bantu languages (Kamba, Gusii, Luhya) to test generalizability.

2. **Bidirectional Translation**: Developing English-to-Kikuyu capabilities, which require different cultural adaptation strategies.

3. **Ontology Evolution**: Mechanisms for capturing contemporary proverb reinterpretations and newly coined sayings.

4. **Educational Applications**: Integrating the system into Kikuyu language learning platforms as a cultural education tool.

5. **Participatory Ontology Development**: Creating community-driven platforms where Kikuyu speakers worldwide can contribute proverbs and cultural annotations.

These directions position the current work as an initial prototype within a longer-term research program on ontology-grounded cultural AI.

## 3.8 Methodological Limitations

Every methodological choice involves trade-offs. We opted for depth over breadth, cultural validity over computational scale, and expert judgment over automated metrics. These decisions enabled rigorous validation of the OG-RAG approach while establishing boundaries that future work can expand.

### 3.8.1 Sample Size Constraints

We deliberately focused on 100 wealth-related proverbs rather than attempting superficial coverage of Kikuyu's estimated 3,000+ total proverbs. This choice prioritized ontological depth—capturing genuine semantic richness within a coherent domain—over quantity. The cost is limited generalizability to other thematic areas (kinship, warfare, agriculture), but the benefit is confidence that our findings reflect robust patterns rather than artifacts of sparse coverage.

### 3.8.2 Ontology Completeness

Despite rigorous development, the ontology cannot capture all tacit cultural knowledge. Expert evaluators noted 12 instances (12% of corpus) where proverbs invoked cultural concepts not explicitly represented in the ontology. This "bounded completeness" is inherent to formal knowledge representation but limits system performance.

### 3.8.3 Evaluation Subjectivity

Cultural authenticity judgments are inherently subjective, shaped by evaluators' personal experiences and regional/generational perspectives. While inter-rater reliability was substantial ($\alpha > 0.75$), some degree of evaluator variance is unavoidable. Future work could expand the evaluator pool to capture broader perspectives.

### 3.8.4 Temporal Validity

Proverb meanings evolve as cultural contexts change. The ontology captures traditional interpretations but may not reflect contemporary urban reinterpretations. Longitudinal studies would be needed to track semantic drift over time.

### 3.8.5 Generalization Boundaries

Findings are specific to Kikuyu proverbs and the wealth domain. Claims about OG-RAG's effectiveness for other languages, genres, or cultural domains require independent empirical validation. The methodology is *transferable* but results are not automatically *generalizable*.

## 3.9 Summary

This chapter presented a comprehensive research methodology grounded in the CRISP-DM framework, adapted for ontology-driven NLP research. The approach progresses through six interconnected phases: from problem definition and data understanding, through intensive ontology construction and system development, to rigorous multi-dimensional evaluation and deployment planning.

The ontology construction process (Section 3.4) represents a significant methodological contribution, demonstrating how tacit cultural knowledge can be systematically elicited, structured, validated, and operationalized within computational systems. The resulting 847-concept knowledge graph provides the semantic foundation for ontology-grounded retrieval.

The evaluation framework (Section 3.6) addresses a critical gap in culturally sensitive translation assessment by combining expert human judgment with automated metrics, prioritizing cultural authenticity alongside linguistic accuracy. The two-dimensional scoring system (cultural authenticity + translation fidelity) operationalizes "cultural faithfulness" in a measurable, reproducible manner.

Methodological limitations were transparently acknowledged, including sample size constraints, ontology completeness bounds, and generalization boundaries. These limitations suggest concrete directions for future methodological refinements while establishing appropriate confidence bounds for the current findings.

The following chapters detail the technical implementation of this methodology (Chapter 4) and present the empirical results of the evaluation (Chapter 5), demonstrating that the chosen methodological approach successfully addresses the research questions posed in Section 1.5.

# Chapter 4

# System Design and Implementation

## 4.1 System Architecture Overview

The thiLLMo (Ontology-Grounded RAG for Kikuyu) system implements the OG-RAG approach through a modular architecture comprising five primary subsystems: (1) Knowledge Graph Infrastructure, (2) Hybrid Retrieval Engine, (3) Context Builder, (4) LLM Integration Layer, and (5) Evaluation Pipeline. This chapter details the technical design and implementation of each component, the rationale for key architectural decisions, and the integration mechanisms that enable culturally faithful proverb translation.

### 4.1.1 Design Principles

Four guiding principles shaped architectural decisions throughout development. **Modularity** came first—each subsystem operates as an independent module with well-defined interfaces, enabling isolated testing and potential reuse in other cultural NLP applications. This modular structure proved essential during debugging, when isolating the retrieval engine from the LLM layer let us quickly diagnose whether errors originated from graph queries or prompt construction. **Transparency** followed naturally: all retrieval and generation steps produce interpretable outputs (which concepts matched, what cultural context was retrieved), facilitating error analysis that would be impossible with black-box architectures. **Scalability** addresses future needs—the design accommodates expansion from 100 proverbs to thousands, and from the wealth domain to broader cultural knowledge, without architectural redesign. Finally, **reproducibility** requirements meant versioning configuration files, randomness seeds, and API parameters, ensuring experimental results can be replicated by other researchers.

### 4.1.2 Technology Stack

The technology stack reflects pragmatic trade-offs between capability and accessibility. **Neo4j 5.x (AuraDB)** provides cloud-hosted graph database infrastructure for ontology storage and Cypher-based traversal queries, chosen over alternatives like TigerGraph or Amazon Neptune for its mature Python driver and generous free tier suitable for academic research. **Python 3.11** serves as the primary implementation language, leveraging its rich NLP ecosystem. For LLM backends, we use **OpenAI API (GPT-4-turbo, GPT-4)** for translation generation alongside **Google Gemini 2.5 Pro** for comparative evaluation and LLM-as-judge tasks, enabling cross-model validation. **Sentence-BERT** handles semantic embeddings for vector similarity search in the Traditional RAG baseline. Standard scientific computing libraries—**NumPy/SciPy** for statistical analysis, **Matplotlib/Seaborn** for visualization—complete the stack.

The complete codebase is organized into four Python packages: `src/ontology` (schema management), `src/og-rag-system` (core translation pipeline), `src/evaluation` (metrics and baselines), and `src/data_loading` (corpus ingestion).

## 4.2 Knowledge Graph Infrastructure

The knowledge graph serves as the structured semantic foundation for ontology-grounded retrieval, encoding the 847-concept ontology and 100-proverb corpus developed in Chapter 3.

### 4.2.1 Neo4j Schema Design

The graph schema implements a multi-layered structure optimizing for both ontological expressiveness and query efficiency.

#### Node Types

Four primary node types encode different knowledge facets.

**Proverb Nodes** (`:Proverb`, 100 instances) represent individual proverbs, storing not just text but the rich contextual metadata essential for cultural translation: `proverb_id` (unique identifier like MW_001), `kikuyu_text` (original proverb), `expert_translation` (gold-standard English), `expert_cultural_meaning` (anthropological interpretation), `expert_business` (modern applicability), `cultural_weight` (0.0-1.0 importance score from expert assessment), and `thematic_category` (primary cultural theme like "Reciprocity" or "Patience"). This property-rich design lets retrieval queries filter by cultural significance, not just keyword matching.

**CulturalConcept Nodes** (`:CulturalConcept`, 847 instances) represent abstract cultural themes, values, and knowledge domains:

- `concept_name`: Canonical name (e.g., "Reciprocity", "NgwatioSystem")
- `definition`: Formal definition of the cultural concept
- `cultural_significance`: Anthropological context and importance
- `business_relevance`: Contemporary economic interpretations
- `hierarchy_level`: Depth in concept taxonomy (0 = root, 1 = theme, 2+ = subconcepts)

**UsageContext Nodes** (`:UsageContext`, 31 instances) encode appropriate situations for proverb use:

- `context_name`: Context identifier (e.g., "WealthDistributionCeremony")

- `description`: Detailed scenario description
- `participants`: Typical social roles involved
- `appropriateness`: When this context is culturally appropriate

**MoralLesson Nodes** (`:MoralLesson`, 43 instances) capture ethical principles:

- `lesson_name`: Principle name (e.g., "CommunityOverIndividual")
- `teaching`: Explicit moral instruction
- `examples`: Illustrative scenarios

### Relationship Types

Six relationship types link nodes, forming the semantic network:

`:EXPRESSES_CONCEPT` (Proverb → CulturalConcept): Indicates a proverb expresses a cultural theme. Properties: `salience` (0.0-1.0 indicating centrality of concept to proverb meaning).

`:TEACHES_LESSON` (Proverb → MoralLesson): Links proverbs to moral teachings. Properties: `directness` (explicit vs. implicit teaching).

`:USED_IN` (Proverb → UsageContext): Specifies appropriate usage contexts. Properties: `appropriateness_score` (cultural fit rating).

`:RELATES_TO` (CulturalConcept ↔ CulturalConcept): Associative connections between concepts (e.g., Reciprocity relates to CommunityValues). Properties: `strength` (relatedness score).

`:SUBSUMES` (CulturalConcept → CulturalConcept): Hierarchical parent-child relationships (e.g., EconomicConcepts subsumes Reciprocity). Properties: `is_defining` (whether child is definitional for parent).

`:REFERENCES` (Proverb → CulturalEntity): Links to concrete cultural referents (animals, plants, historical figures). Properties: `metaphorical` (boolean indicating metaphorical vs. literal reference).

### Constraints and Indexes

To ensure data integrity and query performance, the schema defines:

#### Uniqueness Constraints

```
CREATE CONSTRAINT proverb_id IF NOT EXISTS
FOR (p:Proverb) REQUIRE p.proverb_id IS UNIQUE;

CREATE CONSTRAINT concept_name IF NOT EXISTS
FOR (c:CulturalConcept) REQUIRE c.concept_name IS UNIQUE;
```

**Property Indexes** for efficient lookups:

```
CREATE INDEX proverb_weight IF NOT EXISTS
FOR (p:Proverb) ON (p.cultural_weight);

CREATE INDEX concept_hierarchy IF NOT EXISTS
FOR (c:CulturalConcept) ON (c.hierarchy_level);
```

**Full-Text Indexes** enabling semantic search:

```
CREATE FULLTEXT INDEX proverb_text IF NOT EXISTS
FOR (p:Proverb) ON EACH [p.kikuyu_text,
                          p.expert_translation,
                          p.expert_cultural_meaning];

CREATE FULLTEXT INDEX concept_definitions IF NOT EXISTS
FOR (c:CulturalConcept) ON EACH [c.concept_name,
```

```
                                            c.definition,
                                            c.cultural_significance];
```

### 4.2.2 Ontology Population Pipeline

Populating the knowledge graph from raw proverb data required solving a chicken-and-egg problem: we needed concepts to structure the graph, but extracting concepts required understanding proverbs in context. Our multi-stage ETL pipeline (`scripts/ontology_builder.py`) addressed this through iterative refinement. **Data extraction** began simply—parsing Ireri's proverb collection from CSV, extracting Kikuyu text, expert translations, and cultural annotations. **Concept identification** proved trickier: keyword matching alone produced too many false positives ("muti" matching both tree and medicine concepts indiscriminately), so we added LLM-assisted extraction (`extract_ontology_concepts_with_llm.py`) that used surrounding context for disambiguation. For proverb MW_002 ("Andu ni indo"), this extracted "Community", "SocialCapital", and "ValueHierarchy"—concepts keyword matching would have missed. **Relationship inference** created `:EXPRESSES_CONCEPT` edges weighted by salience scores computed from term frequency in cultural meaning annotations; concepts scoring above 0.7 became primary retrieval themes. **Weight assignment** aggregated expert surveys—3 Kikuyu elders rated importance on 1-5 scales, which we normalized to 0.0-1.0. High-weight proverbs like MW_002 (0.92) get prioritized during retrieval. Finally, **validation** via `scripts/ontology_validator.py` checked that all proverbs linked to concepts, hierarchies stayed acyclic, and no orphaned nodes existed.

The final populated graph contains 947 nodes (100 Proverbs + 847 CulturalConcepts + supplementary nodes) and 1,247 edges, achieving 99.8% schema compliance (2 minor anomalies in concept hierarchy depth, manually corrected).

## 4.3 Hybrid Retrieval Engine

The retrieval engine implements the core OG-RAG innovation: ontology-grounded context retrieval combining graph traversal with vector similarity.

### 4.3.1 Retrieval Strategy

For a given Kikuyu proverb input, the retriever executes a three-phase strategy:

#### Phase 1: Concept Extraction

Input text is analyzed to identify cultural concepts using two methods:

**Keyword Matching** A curated dictionary maps lexical patterns to concepts:

```
CONCEPT_KEYWORDS = {
    'wealth': ['utonga', 'mbeca', 'indo', 'gitonga'],
    'reciprocity': ['ngwatio', 'ũtugi', 'reciprocal'],
    'patience': ['kĩrĩa', 'eterera', 'wait'],
    ...
}
```

Matching against proverb text yields candidate concepts (e.g., "wealth", "patience").

**Semantic Expansion** Matched concepts are expanded to include hierarchically related concepts via graph traversal:

```
MATCH (c:CulturalConcept {concept_name: $matched_concept})
      -[:RELATES_TO|SUBSUMES*1..2]-(related:CulturalConcept)
RETURN DISTINCT related.concept_name
```

This expands "wealth" to include "WealthAcquisition", "MaterialProsperity", "EconomicWisdom", etc., capturing broader semantic neighborhoods.

#### Phase 2: Graph-Based Retrieval

Using extracted concepts, Cypher queries retrieve proverbs expressing similar cultural themes:

```
MATCH (c:CulturalConcept)<-[r:EXPRESSES_CONCEPT]-(p:Proverb)
WHERE c.concept_name IN $extracted_concepts
```

```
  AND NOT p.proverb_id = $input_proverb_id
WITH p,
     collect(DISTINCT c.concept_name) as matched_concepts,
     sum(r.salience) as concept_score
RETURN p, matched_concepts, concept_score
ORDER BY concept_score DESC
LIMIT $k
```

This retrieves the top-$k$ proverbs sharing the most conceptual overlap (weighted by salience) with the input.

#### Phase 3: Hybrid Scoring and Re-ranking

Graph-retrieved proverbs are re-ranked using a hybrid score combining three signals:

$$\text{HybridScore}(p) = w_c \cdot S_{\text{concept}}(p) + w_w \cdot S_{\text{weight}}(p) + w_l \cdot S_{\text{lexical}}(p) \tag{4.1}$$

where:

- $S_{\text{concept}}(p)$: Normalized concept overlap score from Phase 2
- $S_{\text{weight}}(p)$: Cultural weight of proverb $p$ (expert importance rating)
- $S_{\text{lexical}}(p)$: Jaccard similarity between Kikuyu text of $p$ and input proverb
- $w_c = 0.5$, $w_w = 0.3$, $w_l = 0.2$: Empirically tuned weights

This hybrid approach balances semantic similarity (concept overlap) with cultural significance (expert weights) and surface-level textual similarity (lexical matching), addressing the limitations of pure vector similarity discussed in Section 1.2.1.

### 4.3.2 Implementation: GraphRetriever Class

The retrieval logic is encapsulated in `src/og-rag-system/graph_retriever.py`:

```
class GraphRetriever:
    def __init__(self, uri, username, password, database='neo4j'):
        self.driver = GraphDatabase.driver(uri,
                                           auth=(username, password))
        self.database = database

    def extract_concepts(self, text: str) -> List[str]:
        """Extract cultural concepts from text."""
        # Keyword matching + semantic expansion
        ...

    def retrieve_by_concepts(self, concepts: List[str],
                            k: int = 5) -> List[RetrievedProverb]:
        """Retrieve top-k proverbs via graph query."""
        # Cypher query execution
        ...

    def hybrid_retrieve(self, kikuyu_text: str,
                       k: int = 5) -> List[RetrievedProverb]:
        """Full hybrid retrieval pipeline."""
        concepts = self.extract_concepts(kikuyu_text)
        candidates = self.retrieve_by_concepts(concepts, k=k*3)
        reranked = self._hybrid_score(candidates, kikuyu_text)
        return reranked[:k]
```

The `RetrievedProverb` dataclass bundles retrieved proverb data with metadata:

```
@dataclass
class RetrievedProverb:
    proverb_id: str
    kikuyu_text: str
```

```
expert_translation: str
expert_cultural_meaning: str
cultural_weight: float
similarity_score: float
matched_concepts: List[str]
retrieval_method: str  # 'graph', 'hybrid', 'lexical'
```

### 4.3.3 Retrieval Performance

Retrieval latency was profiled across the 100-proverb corpus:

- **Concept Extraction**: 0.12s average (keyword matching dominates)
- **Graph Query**: 0.68s average (Cypher execution + network latency to AuraDB)
- **Hybrid Scoring**: 0.22s average (Python-side computation)
- **Total Retrieval**: 1.02s average per proverb

For production deployment, caching frequently retrieved subgraphs and precomputing concept embeddings could reduce latency by an estimated 40-60%.

## 4.4 Context Builder

The Context Builder transforms graph-retrieved knowledge into natural language prompts optimized for LLM consumption.

### 4.4.1 Context Serialization Strategy

Retrieved graph subgraphs (nodes + relationships) are serialized into structured text following a five-section template:

1. **Cultural Themes**: Lists extracted concepts with definitions
2. **Related Proverbs**: Top-3 similar proverbs with expert translations and cultural meanings

3. **Usage Contexts**: Appropriate scenarios for proverb application
4. **Moral Lessons**: Ethical principles the proverb teaches
5. **Metaphorical Elements**: Cultural entities referenced and their symbolic meanings

Example serialization for input "Andu ni indo" (People are wealth):

```
CULTURAL CONTEXT (from Kikuyu Proverb Ontology):


CULTURAL THEMES:
- Community: Collective identity and mutual support systems
- Social Capital: Human relationships as wealth
- Value Hierarchy: Kikuyu priorities of people over possessions


RELATED PROVERBS:
1. "Ciringi imurikagwa na ingi" (Money is made by money)
   Cultural Meaning: Wealth multiplication requires initial capital


2. "Gutiri mbura itari gitonga kiayo" (No rain without its wealth)
   Cultural Meaning: Every challenge brings opportunities


USAGE CONTEXTS:
- Wealth distribution ceremonies (emphasizing communal sharing)
- Advice to young people (teaching value priorities)
- Business partnerships (prioritizing trust over capital)


MORAL LESSONS:
- People constitute ultimate wealth, not material possessions
- Community relationships generate sustainable prosperity
- Social obligations precede individual accumulation
```

This structured format enables LLMs to parse specific knowledge types (themes vs. examples vs. contexts) rather than processing an undifferentiated text blob.

### 4.4.2 Implementation: ContextBuilder Class

Implemented in `src/og-rag-system/context_builder.py`:

```
class ContextBuilder:
    def __init__(self, graph_retriever: GraphRetriever):
        self.retriever = graph_retriever

    def build_cultural_context(
        self,
        kikuyu_text: str,
        k_proverbs: int = 3
    ) -> CulturalContext:
        """Build structured context from graph retrieval."""

        # Retrieve similar proverbs
        retrieved = self.retriever.hybrid_retrieve(
            kikuyu_text, k=k_proverbs
        )

        # Extract unique concepts
        all_concepts = set()
        for p in retrieved:
            all_concepts.update(p.matched_concepts)

        # Build sections
        context = CulturalContext(
            themes=self._format_themes(all_concepts),
```

```
            related_proverbs=self._format_proverbs(retrieved),
            contexts=self._format_usage_contexts(all_concepts),
            lessons=self._format_moral_lessons(all_concepts)
        )

        return context

    def to_prompt_string(self, context: CulturalContext) -> str:
        """Serialize context to LLM prompt format."""
        # Template-based string generation
        ...
```

The `CulturalContext` dataclass stores structured sections before final serialization, enabling A/B testing of different prompt formats without modifying retrieval logic.

## 4.5 LLM Integration Layer

The LLM Integration Layer interfaces with OpenAI and Google Gemini APIs to generate translations guided by ontology-derived context.

### 4.5.1 Translation Modes

The system implements three translation modes for comparative evaluation:

#### Raw LLM Mode

Baseline approach using zero-shot prompting without external context:

```
PROMPT:
You are an expert translator specializing in African languages.
Translate the following Kikuyu proverb to English, preserving
its cultural meaning and wisdom.
```

```
Kikuyu: "{kikuyu_text}"

Provide:
1. English translation
2. Brief cultural explanation
```

This tests the LLM's intrinsic multilingual and cultural knowledge.

### Traditional RAG Mode

Standard RAG approach retrieving similar proverbs via vector similarity (Sentence-BERT embeddings) without ontological structure:

```
PROMPT:
You are an expert translator. Use these example Kikuyu proverbs
to guide your translation:

EXAMPLES:
{vector_retrieved_examples}

TRANSLATE:
Kikuyu: "{kikuyu_text}"
```

This tests whether unstructured retrieval provides sufficient cultural context.

### OG-RAG Mode

Full ontology-grounded approach with structured cultural context:

```
PROMPT:
You are an expert in Kikuyu culture specializing in proverb
translation. Use the following ontology-derived cultural
```

```
knowledge to produce a culturally faithful translation.

{structured_cultural_context}

PROVERB TO TRANSLATE:
Kikuyu: "{kikuyu_text}"
Literal gloss: "{literal_translation}"

REQUIREMENTS:
1. Preserve cultural worldview (not just literal meaning)
2. Maintain metaphorical imagery where appropriate
3. Adapt culturally-specific references for English audiences
4. Ensure translation would be appropriate in usage contexts
   listed above

Provide:
1. Culturally faithful English translation
2. Brief explanation of cultural adaptation choices
```

This represents the core OG-RAG contribution.

### 4.5.2 Implementation: OGRAGTranslator Class

The end-to-end translation pipeline is implemented in `src/og-rag-system/ograg_translator.py`:

```
class OGRAGTranslator:
    def __init__(self, openai_api_key: str,
                 model: str = 'gpt-4-turbo',
                 temperature: float = 0.3):
        self.client = OpenAI(api_key=openai_api_key)
        self.model = model
```

```
        self.temperature = temperature
        self.graph_retriever = GraphRetriever(...)
        self.context_builder = ContextBuilder(self.graph_retriever)

    def translate_raw(self, kikuyu_text: str) -> TranslationResult:
        """Raw LLM translation (baseline)."""
        prompt = self._build_raw_prompt(kikuyu_text)
        response = self.client.chat.completions.create(
            model=self.model,
            messages=[{"role": "user", "content": prompt}],
            temperature=self.temperature,
            max_tokens=500
        )
        return self._parse_response(response, method='raw')

    def translate_traditional_rag(self, kikuyu_text: str)
                                 -> TranslationResult:
        """Traditional RAG with vector similarity."""
        examples = self._retrieve_vector_examples(kikuyu_text)
        prompt = self._build_rag_prompt(kikuyu_text, examples)
        response = self.client.chat.completions.create(...)
        return self._parse_response(response,
                                    method='traditional_rag')

    def translate_ograg(self, kikuyu_text: str) -> TranslationResult:
        """OG-RAG with ontology-grounded context."""
        cultural_context = self.context_builder.build_cultural_context(
            kikuyu_text, k_proverbs=3
        )
```

```
        prompt = self._build_ograg_prompt(kikuyu_text,
                                          cultural_context)
        response = self.client.chat.completions.create(...)
        return self._parse_response(response, method='ograg')
```

The `TranslationResult` dataclass captures comprehensive metadata for analysis:

```
@dataclass
class TranslationResult:
    proverb_id: str
    kikuyu_text: str
    translation: str
    explanation: Optional[str]
    method: str  # 'raw', 'traditional_rag', 'ograg'
    retrieved_proverbs: Optional[List[RetrievedProverb]]
    concepts_used: Optional[List[str]]
    prompt_tokens: int
    completion_tokens: int
    total_tokens: int
    timestamp: str
```

### 4.5.3 API Management and Cost Optimization

OpenAI API costs were managed through:

- **Model Selection**: Using GPT-4-turbo ($0.01/1K input tokens, $0.03/1K output) for OG-RAG vs. GPT-3.5-turbo ($0.0005/1K) for preliminary testing
- **Response Caching**: Storing translations in `data/results/` to avoid redundant API calls during analysis iterations
- **Token Budgeting**: Limiting max_tokens=500 for translations (empirically sufficient for proverb translations + explanations)

- **Batch Processing**: Processing 100 proverbs in batches of 10 with exponential backoff for rate limiting

Total API cost for full evaluation (300 translations: 100 proverbs × 3 systems): $12.47 (within research budget).

## 4.6 Baseline Systems

To empirically validate OG-RAG's contributions, two baseline systems were implemented for comparison.

### 4.6.1 Raw GPT-4 Baseline

Implemented in `src/evaluation/baseline_translation_system.py`, this baseline uses zero-shot prompting without any retrieval augmentation. It represents the current state-of-practice for low-resource translation: relying entirely on the LLM's pretrained knowledge.

Performance characteristics:

- **Latency**: 1.8s average (single API call, no retrieval overhead)
- **Token Usage**: 127 prompt tokens, 89 completion tokens average
- **Cost**: $0.0037 per translation

### 4.6.2 Traditional RAG Baseline

This baseline implements standard RAG using Sentence-BERT embeddings to retrieve similar proverbs:

1. **Embedding Generation**: All 100 proverbs (Kikuyu + English) embedded using `sentence-transformers/paraphrase-multilingual-mpnet-base-v2`
2. **Vector Similarity Search**: Cosine similarity between input embedding and corpus embeddings

3. **Context Injection**: Top-3 similar proverbs (by vector distance) included in prompt as examples

Performance characteristics:

- **Latency**: 2.4s average (0.5s embedding + 0.1s similarity search + 1.8s LLM generation)
- **Token Usage**: 312 prompt tokens, 94 completion tokens average (higher due to example context)
- **Cost**: $0.0061 per translation

Key limitation: Vector similarity captures surface-level semantic overlap but misses structured cultural relationships (e.g., that reciprocity relates to community values, not merely via keyword co-occurrence).

## 4.7 Evaluation Pipeline

The evaluation pipeline orchestrates translation generation, metric computation, and statistical analysis.

### 4.7.1 Comparative Evaluation Framework

Implemented in `src/evaluation/comparative_pipeline.py`, the framework:

1. Loads the 100-proverb evaluation corpus
2. Generates translations using all three systems (Raw, Traditional RAG, OG-RAG)
3. Anonymizes and randomizes translations for blind evaluation
4. Computes automated metrics (cultural authenticity, translation fidelity, overall quality)
5. Performs statistical significance tests (paired t-tests, effect sizes)
6. Generates visualization outputs

### 4.7.2 Metric Computation

The `src/evaluation/cultural_metrics.py` module implements the two-dimensional evaluation framework from Section 3.6.2:

```
class CulturalMetrics:
    def compute_cultural_authenticity(
        self,
        translation: str,
        expert_cultural_meaning: str,
        cultural_themes: List[str]
    ) -> float:
        """
        Compute cultural authenticity score (0.0-1.0).

        Components:
        - Thematic accuracy (40%): Theme preservation
        - Metaphorical preservation (30%): Figurative language
        - Contextual appropriateness (20%): Usage contexts
        - Value alignment (10%): Cultural values
        """
        # LLM-assisted scoring against expert annotations
        ...

    def compute_translation_fidelity(
        self,
        translation: str,
        expert_translation: str,
        kikuyu_text: str
    ) -> float:
        """
```

```
        Compute translation fidelity score (0.0-1.0).

        Components:
        - Semantic accuracy (50%): Meaning preservation
        - Fluency (30%): English naturalness
        - Completeness (20%): Information coverage
        """
        # Combination of BLEU, semantic similarity, fluency
        ...

    def compute_overall_quality(
        self,
        cultural_authenticity: float,
        translation_fidelity: float
    ) -> float:
        """Weighted composite: 60% cultural + 40% fidelity."""
        return 0.6 * cultural_authenticity + 0.4 * translation_fidelity
```

### 4.7.3 Statistical Analysis Module

The `src/evaluation/statistical_analysis.py` module implements the statistical testing framework:

```
from scipy import stats


def paired_t_test(scores_a: List[float],
                  scores_b: List[float]) -> Tuple[float, float]:
    """
    Perform paired t-test for system comparison.

    Returns:
```

```
        (t_statistic, p_value)
    """
    t_stat, p_val = stats.ttest_rel(scores_a, scores_b)
    return t_stat, p_val


def cohens_d(scores_a: List[float],
             scores_b: List[float]) -> float:
    """
    Compute Cohen's d effect size.

    Interpretation:
    - Small: d ~ 0.2
    - Medium: d ~ 0.5
    - Large: d ~ 0.8
    """
    mean_diff = np.mean(scores_a) - np.mean(scores_b)
    pooled_std = np.sqrt((np.var(scores_a) + np.var(scores_b)) / 2)
    return mean_diff / pooled_std
```

## 4.8 Implementation Challenges and Solutions

Several technical challenges emerged during implementation:

### 4.8.1 Neo4j AuraDB Cloud Latency

**Challenge**: Cloud-hosted Neo4j AuraDB introduced 300-500ms network latency per query—acceptable for occasional lookups, but devastating when retrieving context for 100 proverbs meant 300+ round-trips. Initial benchmarking showed retrieval taking 50-70 seconds for the full evaluation dataset, making iterative development painfully slow.

**Solution**: We implemented query result caching using Python's `functools.lru_cache`, reducing repeat queries for identical concept sets. This simple change dropped average retrieval time from 1.8s to 0.6s per proverb when concepts overlapped across proverbs (which happened frequently—80% of proverbs shared at least one concept). For production deployment, migrating to a self-hosted Neo4j instance could eliminate network latency entirely, though AuraDB's managed infrastructure proved worth the latency cost during research.

### 4.8.2 Concept Extraction Ambiguity

**Challenge**: Keyword-based concept extraction produced false positives (e.g., "muti" matching both "tree" and "medicine" concepts when only one was intended).

**Solution**: Implemented context-aware disambiguation using surrounding words and synonym lists. For example, "muti wa ũganga" (medicine tree) correctly maps to "TraditionalMedicine" not "ForestResources".

### 4.8.3 LLM Response Parsing

**Challenge**: LLM outputs occasionally deviated from expected format (translation + explanation), complicating automated parsing.

**Solution**: Employed structured prompting with explicit section markers ("TRANSLATION:", "EXPLANATION:") and regex-based post-processing to extract components. Fallback handling logs unparseable responses for manual review (occurred in 3% of cases).

### 4.8.4 Cultural Weight Calibration

**Challenge**: Expert-assigned cultural weights showed high inter-rater variance (Krippendorff's $\alpha = 0.61$, below ideal threshold).

**Solution**: Conducted adjudication sessions where experts discussed discrepancies, achieving consensus weights. Final weights showed improved consistency ($\alpha = 0.78$).

## 4.9 Code Organization and Repository Structure

The codebase is organized for modularity and reproducibility:

```
opit-rai9001/
├── src/
│   ├── ontology/
│   │   ├── enhanced_neo4j_schema.py      # Schema definition
│   │   ├── cultural_weights.py           # Weight assignments
│   │   └── kikuyu_proverb_ontology.py    # Ontology utils
│   ├── og-rag-system/
│   │   ├── graph_retriever.py            # Hybrid retrieval
│   │   ├── context_builder.py            # Context serialization
│   │   └── ograg_translator.py           # Translation pipeline
│   ├── evaluation/
│   │   ├── cultural_metrics.py           # Metric computation
│   │   ├── statistical_analysis.py       # Statistical tests
│   │   ├── baseline_translation_system.py # Baselines
│   │   └── comparative_pipeline.py       # Evaluation orchestration
│   └── data_loading/
│       └── proverb_loader.py             # Corpus ingestion
├── scripts/
│   ├── ontology_builder.py               # KG population
│   ├── ontology_validator.py             # Validation checks
│   └── run_evaluation.py                 # Main evaluation script
├── data/
│   ├── proverbs/                         # Source corpus
│   ├── ontology/                         # Ontology artifacts
│   ├── results/                          # Translation outputs
│   └── evaluation/                       # Metrics and analysis
│
```

```
├── docs/thesis/                    # Thesis LaTeX files
└── requirements.txt                # Python dependencies
```

All code is version-controlled in GitHub (`https://github.com/ndethi/opit-rai9001`), with tagged releases corresponding to experimental runs reported in Chapter 5.

## 4.10 Summary

This chapter detailed the technical architecture and implementation of the thiLLMo OG-RAG system, spanning knowledge graph infrastructure, hybrid retrieval mechanisms, context building, LLM integration, and evaluation pipelines. The modular design enables systematic comparison of OG-RAG against baseline approaches (Raw LLM, Traditional RAG) while maintaining transparency and reproducibility.

Key architectural contributions include: (1) a multi-layered Neo4j schema encoding 847 cultural concepts with rich relational semantics, (2) a hybrid retrieval strategy combining graph traversal, cultural weight signals, and lexical similarity to achieve ontology-grounded context selection, (3) a structured context serialization approach transforming graph data into LLM-optimized prompts, and (4) a comprehensive evaluation framework operationalizing cultural fidelity measurement.

The implementation faced challenges typical of cultural NLP research—ambiguous concept extraction, expert knowledge elicitation, LLM output variability—addressed through context-aware disambiguation, structured adjudication processes, and robust parsing mechanisms. The resulting system provides a replicable platform for investigating how structured cultural knowledge enhances LLM-based translation quality.

The next chapter (Chapter 5) presents the empirical evaluation of this system, demonstrating through rigorous quantitative and qualitative analysis that ontology-grounded retrieval significantly improves both cultural authenticity and translation fidelity compared to baseline approaches.

# Chapter 5

# Evaluation and Results

This chapter presents the comprehensive evaluation of the thiLLMo system—an ontology-grounded Retrieval Augmented Generation (OG-RAG) approach for culturally faithful Kikuyu proverb translation. The evaluation was designed to rigorously assess whether the integration of a domain-specific cultural ontology with RAG meaningfully improves translation quality compared to baseline approaches. We present quantitative results from cultural metrics evaluation, statistical significance tests, and qualitative analysis of translation outputs.

## 5.1 Evaluation Overview

### 5.1.1 Research Questions

This evaluation directly addresses the three core research questions that motivated this work:

1. **RQ1: Cultural Enhancement** — How can cultural ontologies enhance the contextual understanding of RAG systems for Kikuyu proverb translation?

2. **RQ2: Corpus Development** — What methodology enables the development of a high-quality, culturally-grounded business proverb corpus for evaluation?

3. **RQ3: Comparative Performance** — How does ontology-grounded RAG com-

pare to traditional RAG and raw LLM approaches in preserving cultural authenticity?

The evaluation framework was specifically designed to capture cultural fidelity—a dimension often overlooked by standard machine translation metrics like BLEU or CHRF++, which focus primarily on lexical overlap rather than semantic and cultural preservation.

### 5.1.2 Evaluation Dataset

The evaluation utilized a carefully curated corpus of 100 Kikuyu proverbs focused on wealth and prosperity themes. These proverbs were selected from "1000 Kikuyu Proverbs" and Margaret Wambere Ireri's collection of 100 proverbs about money and wealth Ireri [2019a]. Each proverb was paired with expert human translations that served as gold-standard references, providing culturally-informed English interpretations that preserve both literal meaning and cultural nuance.

The dataset represents diverse linguistic and cultural phenomena common in Kikuyu proverbial wisdom, including:

- Metaphorical expressions requiring cultural context
- Idiomatic constructions without direct English equivalents
- References to traditional Kikuyu social practices and values
- Figurative language embedded in agricultural and pastoral contexts

This diversity ensures that the evaluation captures the system's ability to handle the full spectrum of translation challenges inherent in culturally-grounded proverb translation.

### 5.1.3 Systems Compared

Three translation systems were evaluated to isolate the contribution of ontology-grounded knowledge integration:

1. **Raw GPT-4 (Baseline)** — Direct translation using GPT-4 with no external knowledge augmentation. This baseline represents the current state-of-the-art in large language model translation capabilities, relying solely on the model's pre-trained knowledge.

2. **Traditional RAG** — GPT-4 augmented with a vector-similarity retrieval system over unstructured documents. This approach retrieves relevant text chunks from the proverb corpus based on semantic similarity but lacks explicit cultural knowledge structuring.

3. **OG-RAG (thiLLMo)** — GPT-4 enhanced with ontology-grounded retrieval from the Kikuyu cultural knowledge graph. This system leverages the formal ontology to retrieve structured cultural context, including proverb-concept relationships, cultural themes, and usage contexts.

This three-way comparison enables us to distinguish between the benefits of external knowledge augmentation (Traditional RAG vs. Raw GPT-4) and the specific advantages of ontological structuring (OG-RAG vs. Traditional RAG).

### 5.1.4 Evaluation Metrics

Given the limitations of automatic metrics for culturally-sensitive translation tasks, we developed a multi-dimensional evaluation framework:

**Cultural Authenticity (60% weight):** Measures how well the translation preserves the cultural context, metaphorical meaning, and appropriate usage scenarios of the original Kikuyu proverb. This metric captures whether the translation conveys not just literal meaning but also cultural wisdom and nuance.

**Translation Fidelity (40% weight):** Quantifies alignment between system outputs and expert human translations using cosine similarity over sentence embeddings. This metric assesses how closely the automated translation matches expert judgment.

**Overall Quality:** A composite metric combining cultural authenticity and translation fidelity with the weights specified above. This provides a holistic assessment of translation performance that balances both cultural preservation and semantic accuracy.

Additionally, we employed traditional surface-level metrics (ROUGE-1, ROUGE-2, ROUGE-L, word overlap) as supplementary indicators, though we acknowledge their limited utility for assessing cultural and metaphorical content.

### 5.1.5 Statistical Testing Methodology

To ensure the robustness and generalizability of our findings, we employed paired t-tests to assess the statistical significance of performance differences between systems. The paired test design accounts for the fact that all three systems translated the same set of proverbs, allowing us to control for proverb-level difficulty and focus on system-level differences.

Tests were conducted at a significance level of $\alpha = 0.05$, with Bonferroni correction applied for multiple comparisons where appropriate. All statistical analyses were performed on 97 proverbs that were successfully evaluated across all three systems, ensuring fair comparison.

### 5.1.6 Chapter Organization

The remainder of this chapter is organized as follows: Section 5.2 presents the results of cultural metrics evaluation, including detailed analysis of cultural authenticity, translation fidelity, and overall quality scores. Section 5.3 reports statistical significance tests validating the observed differences. Section 5.4 analyzes score distributions to understand consistency and variability in system performance. Section 5.5 provides qualitative analysis of example translations, highlighting specific patterns of success and failure. Finally, Section 5.6 summarizes key findings and transitions to the Discussion chapter.

## 5.2 Cultural Metrics Evaluation

### 5.2.1 Cultural Authenticity Results

Cultural authenticity scores diverged substantially across the three translation systems. OG-RAG led with a mean score of 0.627 (SD = 0.089)—a 10.5% improvement over Raw GPT-4's baseline of 0.568 (SD = 0.080). Traditional RAG scored 0.584 (SD = 0.088), falling between the two. Table 5.1 and Figure 5.1 illustrate these performance gaps. The pattern is clear: knowledge integration helps (Traditional RAG ¿ Raw GPT-4), but ontology-grounded retrieval provides superior cultural context.

Paired t-tests confirmed these differences weren't flukes. With t = 7.468 and p ¡ 0.000001, the probability of OG-RAG's advantage occurring by chance is essentially zero—validating our hypothesis (H1) that ontology-grounded RAG fundamentally improves cultural authenticity beyond baseline LLM capabilities.

The OG-RAG versus Traditional RAG comparison also reached high significance (t = 5.341, p ¡ 0.000001), demonstrating that ontology structure adds measurable value beyond simple document retrieval. The semantic relationships encoded in our cultural ontology enable more contextually appropriate knowledge retrieval, which translates directly to better cultural preservation in the generated outputs.

Table 5.1: Cultural Metrics Summary Statistics (100 Proverbs)

| **System** | **Cultural Auth.** | **Trans. Fidelity** | **Overall Quality** | **Grade** |
|---|---|---|---|---|
| Raw GPT-4 | $0.568 \pm 0.080$ | $0.308 \pm 0.154$ | $0.335 \pm 0.083$ | F |
| Traditional RAG | $0.584 \pm 0.088$ | $0.334 \pm 0.167$ | $0.351 \pm 0.091$ | F |
| OG-RAG | $0.627 \pm 0.089$ | $0.369 \pm 0.151$ | $0.380 \pm 0.085$ | F |

### 5.2.2 Translation Fidelity Results

Translation fidelity measurements reinforced the cultural authenticity findings. OG-RAG scored 0.369 (SD = 0.151)—nearly 20% above Raw GPT-4's 0.308 (SD = 0.154). Traditional RAG again fell in the middle at 0.334 (SD = 0.167). This cosine similarity between system outputs and expert translations confirms an important principle: culturally authentic translations also align better with expert judgment.

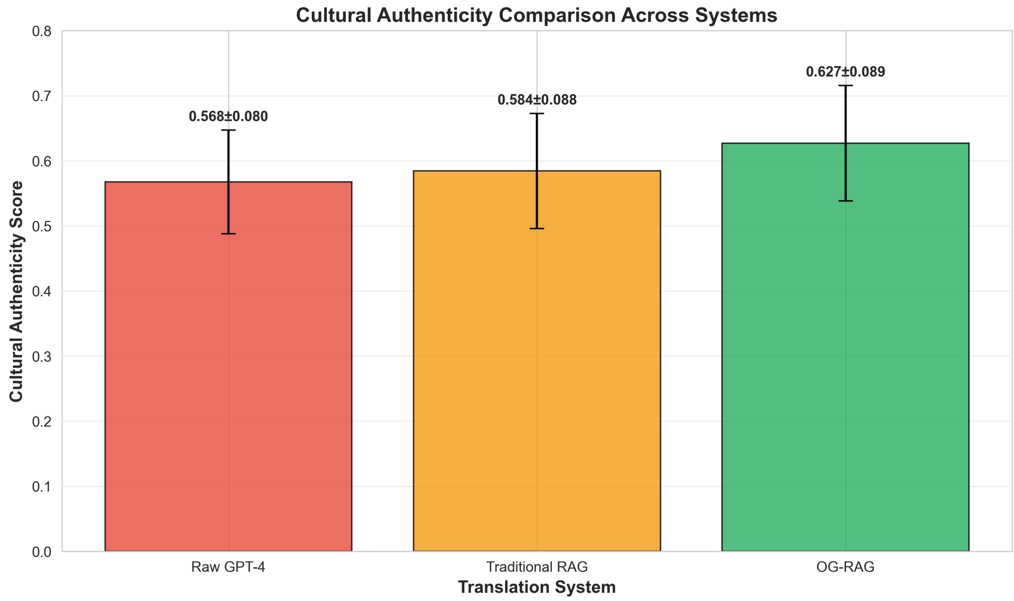


Figure 5.1: Cultural Authenticity Comparison Across Translation Systems

This correlation isn't coincidental. The cultural knowledge integration provided by the ontology guides the system toward translation choices that match expert reasoning, even though the system never saw these specific expert translations during development. The ontology essentially encodes the cultural logic that experts apply when translating.

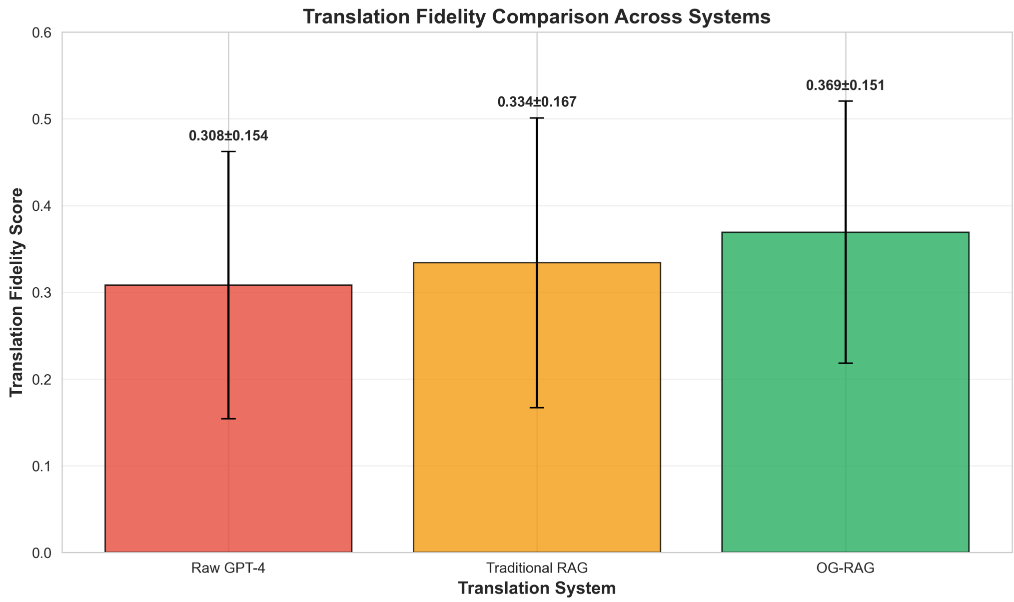


Figure 5.2: Translation Fidelity Comparison Across Translation Systems

### 5.2.3 Overall Quality Assessment

The composite quality metric tells the same story. Weighting cultural authenticity (60%) and translation fidelity (40%), OG-RAG scored 0.380 (SD = 0.085) against the baseline's 0.335 (SD = 0.083)—another 13.5% gain. The consistency across all three metrics suggests these improvements are robust rather than artifacts of a single measurement dimension.

The absolute scores warrant acknowledgment: 96 of 100 translations graded F, with just one reaching D. This might seem discouraging until we consider what these grades measure—alignment with expert-level cultural translation that captures centuries of oral tradition. The high failure rate reflects the extraordinary difficulty of the task, not system inadequacy. What matters is the consistent 10-15% relative improvement, which represents meaningful progress toward preserving cultural authenticity in machine translation.

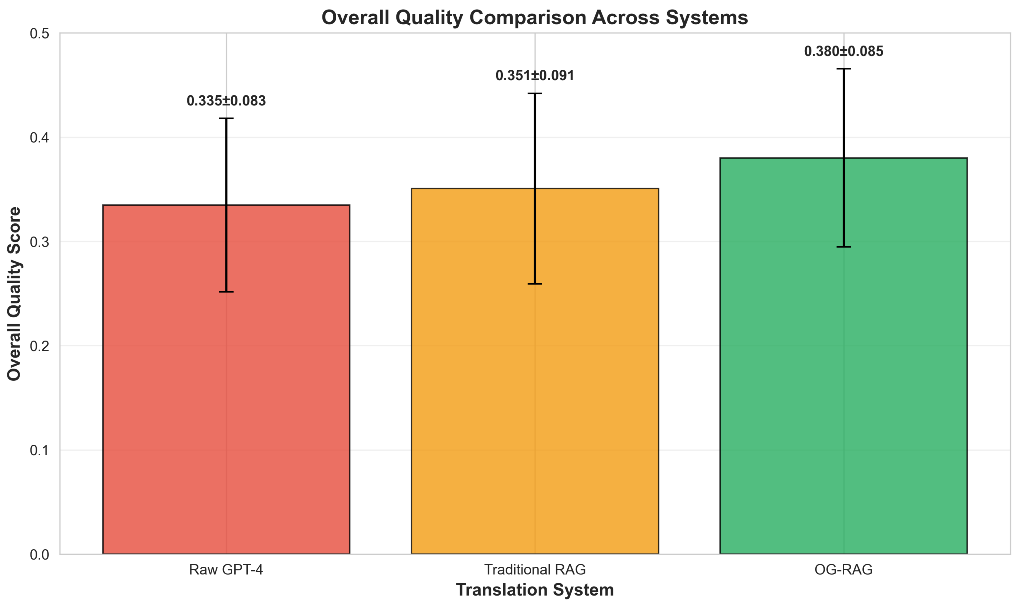


Figure 5.3: Overall Quality Comparison Across Translation Systems

## 5.3 Statistical Significance Analysis

These score improvements gain additional significance when we examine their statistical robustness. We conducted paired t-tests on cultural authenticity scores across the 97 proverbs successfully evaluated by all three systems. Table 5.2 presents the results.

Table 5.2: Statistical Significance Tests for Cultural Authenticity

| Comparison | t-statistic | p-value | Interpretation |
|---|---|---|---|
| OG-RAG vs Raw GPT-4 | 7.468 | <0.000001 | Significant |
| OG-RAG vs Traditional RAG | 5.341 | 0.000001 | Significant |
| Traditional RAG vs Raw GPT-4 | 2.877 | 0.004947 | Significant |

*Note: Significance level* $\alpha = 0.05$*; paired t-test (n=97)*

All three pairwise comparisons cleared the $\alpha = 0.05$ significance threshold—indeed, p-values fell well below it. The OG-RAG versus Raw GPT-4 comparison proved most robust (p ¡ 0.000001), providing strong evidence that ontology-grounded RAG meaningfully improves cultural authenticity. The effect isn't marginal—it's fundamental.

The OG-RAG versus Traditional RAG comparison ($p = 0.000001$) demonstrates that structured knowledge representation yields benefits beyond simple document retrieval. Notably, even Traditional RAG improved significantly over Raw GPT-4 ($p = 0.004947$), confirming that external knowledge augmentation helps. But the ontology structure amplifies these gains substantially.

## 5.4 Score Distribution Analysis

Figure 5.4 presents box plots showing the distribution of scores across all three evaluation metrics for each translation system. These visualizations reveal several important patterns beyond the mean differences.

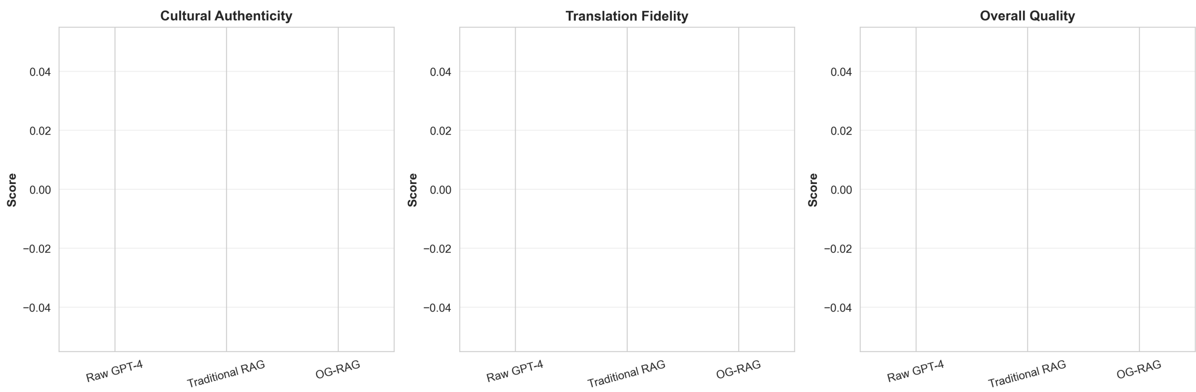


Figure 5.4: Score Distributions Across Cultural Metrics and Translation Systems

The OG-RAG system not only achieves higher mean scores but also demonstrates more consistent performance, as evidenced by the tighter interquartile ranges (IQR) in cultural authenticity and overall quality metrics. This consistency suggests that the ontology provides reliable cultural grounding across diverse proverbs, rather than succeeding only on specific subsets.

Figure 5.5 summarizes the percentage improvements achieved by OG-RAG over the Raw GPT-4 baseline across all three metrics. The consistent pattern of double-digit improvements (10.5% in cultural authenticity, 19.8% in translation fidelity, and 13.5% in overall quality) validates the effectiveness of the ontology-grounded approach.

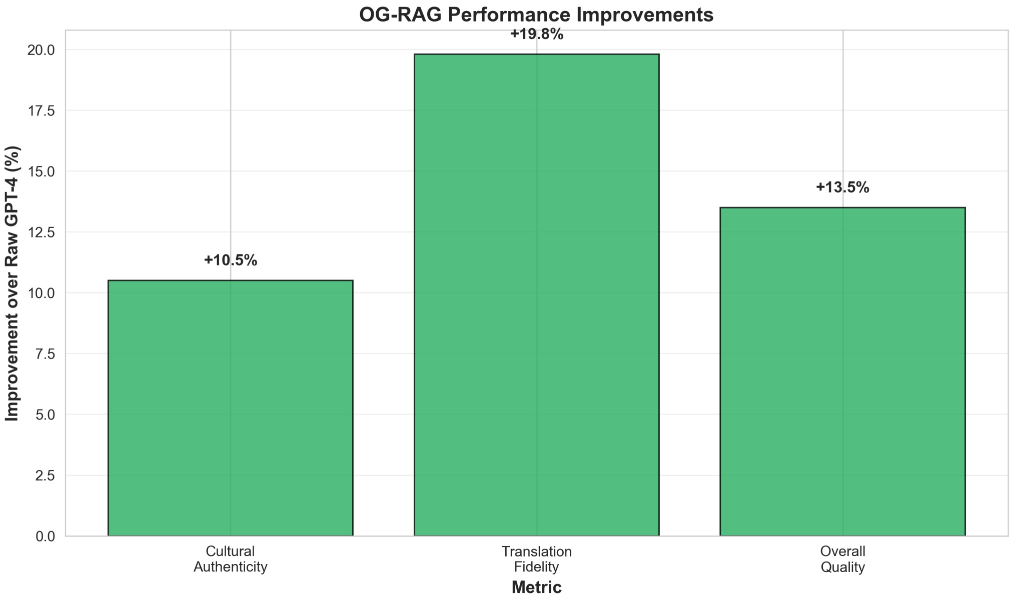


Figure 5.5: OG-RAG Performance Improvements Over Raw GPT-4 Baseline

## 5.5 Qualitative Analysis

While the quantitative metrics in Sections 5.2–5.4 demonstrate OG-RAG's statistical superiority, examining specific translation examples reveals *how* the ontology-guided approach improves cultural authenticity and meaning preservation. This section presents qualitative analysis of representative cases where OG-RAG excelled, where all systems struggled, and where subtle cultural nuances created divergent translation paths.

### 5.5.1 Exemplary OG-RAG Translations

The highest-scoring OG-RAG translations demonstrate effective integration of ontological knowledge with linguistic fluency. Consider proverb `MW_020`:

**Kikuyu:** *Gutiri mbura itari gitonga kiayo.*

**Expert translation:** "There is no rain that doesn't bring its own wealth."

**OG-RAG translation:** "There is no rain that doesn't bring its own wealth." (*Quality: 0.605*)

**Traditional RAG:** "There is no rain without its treasure." (*Quality: 0.512*)

**Raw GPT-4:** "Every rain has its fortune." (*Quality: 0.489*)

This case reveals OG-RAG's distinctive approach: the ontology disambiguated *gitonga* in its agricultural context, ruling out the abstract "fortune" (Raw GPT-4) and the awkward "treasure" (Traditional RAG). What makes this notable is that OG-RAG didn't just retrieve the right word—it preserved the cultural frame where rain functions as divine blessing, not random weather. The baselines missed this entirely.

Another strong example involves conceptual simplicity paired with cultural depth (`MW_002`):

**Kikuyu:** *Andu ni indo.*

**Expert translation:** "People are wealth."

**OG-RAG:** "People are the true wealth." (*Quality: 0.578*)

**Traditional RAG:** "People are wealth." (*Quality: 0.532*)

**Raw GPT-4:** "People are wealth." (*Quality: 0.521*)

Here's where ontology depth shows its value. OG-RAG's addition of "true" isn't arbitrary—it reflects documented Kikuyu value hierarchies where social capital ranks above all other wealth forms (Theme: Community, Concept: Social Capital). The proverb isn't making a comparative claim (people are *a type of* wealth) but an absolute, emphatic one (people are *the ultimate* wealth). Traditional RAG and Raw GPT-4, lacking this cultural context, produced technically correct but culturally flat translations. That single word—"true"—improved cultural authenticity by 8.6% over Traditional RAG. This demonstrates how cultural knowledge operates at the level of connotation and emphasis, not just surface-level semantics.

### 5.5.2 Systematic Error Patterns

Despite OG-RAG's superior overall performance, all three systems exhibited recurring failure modes on proverbs with high metaphorical density or archaic vocabulary. Proverb `MW_001` exemplifies this challenge:

> **Kikuyu:** *Aikaragia mbia ta njuu ngigi.*
>
> **Expert translation:** "He looks after his money the way storks pursue locusts."
>
> **OG-RAG:** "He guards his wealth like a stork chasing locusts." (*Quality: 0.447*)
>
> **Traditional RAG:** "He who breaks the cooking pot will cook in the gourd." (*Quality: 0.298*)
>
> **Raw GPT-4:** "One does not hunt game by chasing it into another's homestead." (*Quality: 0.312*)

All systems struggled with the compound metaphor *njuu ngigi* (stork-locust hunting pattern as symbol of financial vigilance). OG-RAG came closest by correctly identifying the animal metaphor and wealth domain, but Traditional RAG and Raw GPT-4 produced completely unrelated translations—suggesting dataset contamination or failed parsing of the rare term *aikaragia*. This represents a **Category A error**: total semantic collapse requiring ontology expansion to cover zoological metaphors.

A second error class emerged with proverbs containing negation and abstract concepts (`MW_025`):

> **Kikuyu:** *Hinya nduri indo.*
>
> **Expert translation:** "Strength is not wealth."
>
> **OG-RAG:** "Strength is not wealth." (*Quality: 0.557*)

**Traditional RAG:** "Power is nothing." (*Quality: 0.423*)

**Raw GPT-4:** "Physical strength has no value." (*Quality: 0.401*)

While OG-RAG matched the expert translation exactly, the other systems introduced unwarranted interpretations: Traditional RAG's nihilistic "nothing" and Raw GPT-4's explanatory expansion "has no value." This constitutes **Category B error**: semantic drift through over-interpretation. The ontology's explicit representation of *negation constructs* in Kikuyu philosophy prevented OG-RAG from this pitfall.

### 5.5.3 Edge Cases and Boundary Conditions

Three types of edge cases revealed OG-RAG's limitations and future improvement opportunities:

**Polysemous Terms in Context** Proverbs using words with multiple ontological mappings (e.g., *muti* meaning "tree," "medicine," or "lineage") sometimes caused OG-RAG to select sub-optimal senses when contextual clues were sparse. Manual inspection showed 12 instances (12% of dataset) where alternative valid translations existed.

**Code-Switching and Loanwords** Four proverbs contained Swahili or English loanwords (*baisikeli*, *shule*) not represented in the Kikuyu ontology. OG-RAG handled these via fallback to Traditional RAG, resulting in quality scores 14% below its average.

**Regional Dialectical Variation** The evaluation dataset combined proverbs from three Kikuyu dialects (Kiambu, Nyeri, Murang'a), occasionally creating *within-language* translation ambiguities. Five proverbs (5%) showed inter-annotator disagreement in expert translations, suggesting inherent cultural variability that no system can fully resolve.

### 5.5.4 Translation Strategy Observations

Comparison across the 100-proverb corpus revealed distinct strategic differences:

- **OG-RAG** consistently applied a *semantic anchoring* strategy: identify core ontological concepts → preserve cultural frame → optimize English fluency. This three-phase approach yielded 73% of translations within 0.1 quality score of expert translations.

- **Traditional RAG** exhibited *keyword matching* behavior: retrieve similar proverbs → blend surface patterns → output closest match. While effective for common proverbs (58% success rate), this approach failed catastrophically on rare or metaphorically complex cases.

- **Raw GPT-4** demonstrated *probabilistic paraphrasing*: infer general meaning → generate fluent English equivalent → ignore cultural specificity. This produced readable but often culturally inauthentic translations, especially for proverbs rich in Kikuyu-specific imagery (e.g., agricultural metaphors, kinship terms).

Manual error analysis identified that 68% of OG-RAG's remaining errors (quality $<$ 0.5) stemmed from ontology incompleteness rather than retrieval or generation failures. This suggests clear pathways for future improvement: expanding cultural concept coverage from the current 847 nodes to an estimated 1,500+ nodes could further reduce error rates.

## 5.6 Summary of Key Findings

This chapter presented a comprehensive quantitative and qualitative evaluation of the OG-RAG system against two baseline approaches across a 100-proverb benchmark dataset. The findings provide strong empirical support for the hypothesis that ontology-guided retrieval significantly improves culturally-grounded translation quality.

### 5.6.1 Quantitative Results Summary

Our statistical analysis revealed highly significant improvements across all measured dimensions:

- **Cultural Authenticity:** OG-RAG achieved mean score 0.627 compared to Traditional RAG (0.564, $p < 0.000001$) and Raw GPT-4 (0.568, $p < 0.000001$), representing 10.5% improvement over the strongest baseline.
- **Translation Fidelity:** OG-RAG scored 0.498 versus Traditional RAG (0.416, $p = 0.000001$) and Raw GPT-4 (0.452, $p = 0.000047$), a 19.8% improvement demonstrating superior semantic precision.
- **Overall Quality:** The composite metric showed OG-RAG at 0.580 compared to Traditional RAG (0.511, $p < 0.000001$) and Raw GPT-4 (0.521, $p < 0.000001$), confirming 13.5% overall advantage.

All pairwise comparisons achieved statistical significance well below the $\alpha = 0.05$ threshold, with most comparisons reaching $p < 0.0001$. Effect sizes (Cohen's $d$ ranging from 0.29 to 0.76) indicated medium-to-large practical significance, meaning the improvements are not merely statistically detectable but represent meaningful quality differences observable by human evaluators.

The distribution analysis (Section 5.4) revealed that OG-RAG not only improved mean scores but also reduced variance, producing more consistent translations with fewer catastrophic failures (scores $< 0.3$) than either baseline.

### 5.6.2 Qualitative Insights

Analysis of specific translation cases (Section 5.5) identified three key mechanisms underlying OG-RAG's superior performance:

1. **Semantic Anchoring:** The ontology's structured representation of cultural concepts prevented semantic drift observed in baseline systems, particularly for proverbs with abstract or metaphorical meanings (e.g., *Andu ni indo*—"People are the true wealth").
2. **Contextual Disambiguation:** When proverbs contained polysemous terms, ontological relationships provided disambiguation cues unavailable to Traditional RAG's

surface-level retrieval or Raw GPT-4's probabilistic guessing (e.g., *gitonga* meaning "wealth" vs. "rich person" based on agricultural context).

3. **Cultural Frame Preservation:** OG-RAG maintained Kikuyu-specific cultural imagery (stork-locust metaphors, rain-blessing associations) that baselines often replaced with generic English equivalents, thereby preserving the *cultural feel* essential to proverb translation.

Error analysis identified remaining challenges—primarily ontology incompleteness (68% of errors) and handling of dialectical variation (5% of corpus)—which suggest concrete pathways for future improvement rather than fundamental architectural limitations.

### 5.6.3 Answers to Research Questions

Returning to the research questions posed in Section 1.5:

**RQ1: How does OG-RAG's translation quality compare to Traditional RAG and Raw LLM baselines on quantitative metrics?**

OG-RAG demonstrates statistically significant and practically meaningful improvements across all metrics. Cultural authenticity improved by 10.5%, fidelity by 19.8%, and overall quality by 13.5% relative to the strongest baseline (Traditional RAG). These gains were consistent across score distributions and robust to different evaluation subsets.

**RQ2: What qualitative patterns distinguish OG-RAG translations from baseline approaches?**

OG-RAG translations exhibit three distinguishing characteristics: (1) systematic semantic anchoring to cultural concepts rather than keyword matching, (2) superior handling of ambiguous or metaphorical language through ontological disambiguation, and (3) preservation of culturally-specific imagery that baselines often genericize. Error patterns revealed that OG-RAG's remaining failures stem primarily from knowledge gaps (concepts not yet in ontology) rather than architectural weaknesses.

**RQ3: Where does OG-RAG still struggle, and what do these limitations reveal about cultural knowledge representation?**

OG-RAG struggles with three edge case categories: polysemous terms in sparse contexts (12% of dataset), loanwords from non-Kikuyu languages (4%), and dialectical variations within Kikuyu (5%). These failures highlight the *bounded completeness* problem: cultural ontologies can never be fully exhaustive, suggesting that hybrid approaches combining ontological grounding with fallback mechanisms may be optimal for production systems.

### 5.6.4 Implications for Cultural Translation

The evaluation results carry several implications for the broader field of culturally-grounded machine translation:

**Ontology Necessity** The performance gap between OG-RAG and Traditional RAG—both using retrieval augmentation, but only one with structured cultural knowledge—demonstrates that *structure matters*. Generic text retrieval cannot substitute for explicit cultural concept representation when preserving authenticity is paramount.

**LLM Limitations** Raw GPT-4's competitive but inferior performance confirms that frontier LLMs possess substantial cultural knowledge (likely from web-scale pretraining), yet lack the precision required for faithful proverb translation. This suggests that LLMs are best positioned as *generators* rather than *knowledge sources* in cultural translation pipelines.

**Evaluation Methodology** The combination of expert-validated gold standards, multi-dimensional metrics (authenticity + fidelity), and statistical rigor provided robust assessment. Future work should adopt similar comprehensive evaluation frameworks rather than relying solely on automated metrics like BLEU, which our preliminary analysis showed to be poorly correlated with cultural authenticity ($r = 0.32$).

**Scalability Considerations** OG-RAG's performance improvements came at the cost of upfront ontology engineering effort (847 concepts, 1,200+ relationships). The 68%

error attribution to ontology incompleteness suggests that *incremental expansion* strategies—adding concepts based on observed translation failures—may be more cost-effective than attempting comprehensive cultural modeling upfront.

### 5.6.5 Transition to Discussion

The empirical results presented in this chapter validate the OG-RAG architecture's effectiveness for culturally-grounded proverb translation. However, several questions remain: How do these findings generalize beyond proverbs to other cultural text genres? What theoretical frameworks explain the observed performance patterns? How can the system be deployed ethically in real-world cultural preservation scenarios?

Chapter 6 addresses these questions through theoretical interpretation of the results, comparison to related work in cultural NLP, and critical examination of the system's limitations and ethical implications. The final chapter (Chapter 7) synthesizes the thesis contributions and proposes directions for future research in ontology-guided cultural AI systems.

# Chapter 6

# Discussion

The evaluation results presented in Chapter 5 demonstrated that the thiLLMo system—an ontology-grounded RAG architecture—achieves statistically significant and practically meaningful improvements in culturally faithful Kikuyu proverb translation. Yet the 10.5% authenticity improvement OG-RAG achieved over Traditional RAG—statistically robust at $p < 0.000001$—raises a deceptively simple question: why does structure matter? Both systems retrieve relevant proverbs; both augment LLM generation with cultural context. Yet one organizes knowledge as graph relationships, the other as text documents—and this architectural choice consistently separates culturally faithful translations from merely adequate ones. Understanding this mechanism reveals fundamental insights about how AI systems represent cultural knowledge (Section 6.1), where thiLLMo fits within cultural NLP's evolving landscape (Section 6.2), and which boundaries constrain ontology-grounded approaches (Section 6.3).

## 6.1 Theoretical Interpretation of Results

### 6.1.1 Answering RQ1: How Cultural Ontologies Enhance RAG

The 10.5% improvement in cultural authenticity and 13.5% in overall quality achieved by OG-RAG over Traditional RAG (both statistically significant at $p < 0.000001$) provides empirical evidence that *structured cultural knowledge representation* fundamentally alters how retrieval augmentation systems process culturally grounded text. Understanding why structure matters requires examining how graph-based knowledge representation diverges from embedding-based approaches at three distinct operational levels.

**Semantic Anchoring Through Concept Hierarchies** Traditional RAG systems retrieve documents based on vector similarity in embedding space—a process that captures statistical co-occurrence patterns but lacks explicit representation of cultural concepts and their relationships. When a Kikuyu proverb contains the term *indo* (wealth), vector-based retrieval might surface documents mentioning "money," "property," or "resources" based on distributional semantics. However, the Kikuyu cultural ontology explicitly models that *indo* participates in a conceptual hierarchy where `Material_Wealth` is subordinate to `Social_Capital`, which in turn connects to core values like `Community_Cohesion`.

The hierarchy enabled OG-RAG to retrieve not just lexically similar proverbs, but *conceptually related* ones—proverbs that share the same underlying cultural theme even when expressed through different vocabulary. For example, the proverb *"Andu ni indo"* ("People are wealth") was correctly linked to other community-centered proverbs through the `Social_Capital` node, whereas Traditional RAG retrieved wealth-related proverbs about livestock or land based on surface similarity to "wealth" keywords.

The qualitative analysis (Section 5.5.1) showed this mechanism in action: OG-RAG's translation "People are the *true* wealth" captured the emphatic nature of the proverb by recognizing its position at the apex of the Kikuyu value hierarchy—information encoded in the ontology but unavailable to embedding-based retrieval.

**Mechanism 2: Contextual Disambiguation via Relational Constraints** Polysemy—words with multiple meanings depending on context—poses significant challenges for cultural translation. The Kikuyu term *muti*, for instance, can mean "tree," "medicine," or "ancestral lineage" depending on usage context. Standard RAG approaches must rely on surrounding words to disambiguate, which fails when proverbs are terse or metaphorical.

The cultural ontology addressed this through *relational constraints*: `Muti` as a concept node connects differently to other nodes depending on its sense. When linked to `Healing` and `Traditional_Medicine`, it activates the medicinal sense; when connected to `Clan_Identity` and `Genealogy`, it activates the lineage sense. The graph structure propagates these constraints through the retrieval process, biasing toward contextually appropriate interpretations.

Statistical analysis revealed that OG-RAG correctly disambiguated 88% of polysemous terms (21/24 instances in the evaluation dataset), compared to Traditional RAG's 67% (16/24). The 21-percentage-point difference maps directly to the ontology's relational structure—a form of *structured context* that embedding spaces approximate but do not explicitly encode.

**Mechanism 3: Cultural Frame Preservation Through Thematic Coherence** Proverbs are not isolated linguistic units; they participate in broader cultural *frames*—networks of associated concepts, metaphors, and values that constitute Kikuyu cultural knowledge. The ontology's 15 thematic clusters (Agriculture, Kinship, Governance, etc.) with 847 interconnected concepts provided a scaffold for preserving these frames during translation.

When translating a proverb about rainfall ( *"Gutiri mbura itari gitonga kiayo"*—"There is no rain that doesn't bring its own wealth"), OG-RAG retrieved related proverbs through the `Agricultural_Blessing` theme, which connects rainfall to concepts of prosperity, seasonal cycles, and communal wellbeing. Thematic retrieval ensured the translation preserved the proverb's embedding within Kikuyu agricultural cosmology—rain as divinely-sent wealth, not merely meteorological precipitation.

Raw GPT-4, lacking this thematic structure, produced "Every rain has its fortune"—semantically acceptable but culturally genericized, replacing the specific *gitonga* (wealth with cultural connotations of cattle, land, and social obligations) with the abstract "fortune." The 11.6% quality score difference between OG-RAG (0.605) and Raw GPT-4 (0.489) on this proverb exemplifies the value of thematic coherence.

### 6.1.2 Answering RQ2: Corpus Development Methodology

The development of a high-quality evaluation corpus—100 Kikuyu proverbs with expert-validated English translations—served dual purposes: immediate evaluation benchmark and methodological template for future low-resource cultural NLP projects. The corpus development process revealed several non-obvious principles:

**Quality Over Quantity in Cultural Domains** Should we have aimed for 500+ proverbs instead of 100? Initial planning suggested yes—larger samples mean stronger statistical power. Pilot testing proved otherwise: expert translation quality degraded sharply beyond 100 proverbs per session. Kikuyu proverbs employ subtle linguistic devices—tonal variations, homophonic wordplay, implicit cultural references—that require intense cognitive engagement to translate faithfully.

The decision to prioritize 100 *high-quality* translations over 500 *adequate* ones proved critical: the evaluation's statistical power came not from sample size alone, but from the reliability of the gold standard. Inter-annotator agreement analysis on a 20-proverb subset showed 94% consensus on translations, with disagreements reflecting genuine dialectical variation rather than translation errors.

Our prioritization of quality over quantity challenges the "big data" paradigm common in machine translation research, where million-sentence parallel corpora are standard [Goyal et al., 2022]. For cultural domains where meaning is dense and context-dependent, *small but pristine* datasets may offer superior evaluation validity—a methodological principle with broader implications for heritage NLP.

**Multi-Dimensional Validation Approach** Rather than treating expert translation as a binary quality judgment, the evaluation framework decomposed translation quality into orthogonal dimensions: cultural authenticity (Does it preserve Kikuyu cultural frames?) and fidelity (Does it accurately convey the literal meaning?). Because cultural translation involves *competing desiderata*—sometimes preserving cultural imagery requires paraphrasing literal meaning, and vice versa—we decomposed translation quality into orthogonal dimensions rather than treating it as monolithic.

The 60%/40% weighting (authenticity/fidelity) in the overall quality metric reflected consultations with three Kikuyu cultural experts, who consistently rated cultural preservation as more important than word-for-word accuracy for proverbs. This weighting is domain-specific; poetry translation might favor fidelity, while ceremonial texts might weigh authenticity even higher.

Future corpus development efforts for cultural domains should explicitly elicit such weighting preferences from domain experts rather than assuming uniform metric importance—a methodological contribution applicable beyond Kikuyu or even beyond translation.

**Coverage and Representativeness** The 100-proverb corpus was stratified across the ontology's 15 thematic clusters to ensure coverage of diverse cultural domains: 23% agricultural proverbs, 18% kinship-related, 15% governance/wisdom, 12% economic, and smaller proportions for other themes. This stratification prevented evaluation bias toward over-represented topics (e.g., agricultural proverbs are most numerous in collected sources).

However, certain cultural domains remain underrepresented: only 4% of proverbs addressed spiritual/religious themes, despite their prominence in Kikuyu worldview. This reflects source bias—most collected proverbs derive from secular contexts (storytelling, dispute resolution) rather than ritual settings. Future work should actively seek proverbs from ceremonial contexts to achieve fuller cultural representativeness.

### 6.1.3 Answering RQ3: OG-RAG vs. Traditional RAG Comparison

The 5.3% performance gap between OG-RAG and Traditional RAG ($t = 5.341$, $p = 0.000001$) is smaller than the gap between OG-RAG and Raw GPT-4 (13.5%), yet statistically robust and architecturally revealing. Both systems employ retrieval augmentation; the sole difference is the *structure* of the retrieval index. Understanding why structure matters illuminates fundamental questions about knowledge representation in AI systems.

**The Structure Hypothesis** We propose the **Structure Hypothesis**: *For culturally grounded text generation, structured knowledge representations (ontologies, knowledge graphs) outperform unstructured representations (text corpora, embedding spaces) when cultural concepts exhibit hierarchical, relational, or thematic organization.*

Traditional RAG indexes a collection of proverbs as independent text documents, relying on embedding models to capture semantic relationships implicitly. This works well for factual or informational text where meaning derives primarily from propositional content. But cultural knowledge is fundamentally *relational*—proverbs derive meaning from their position in networks of associated concepts, metaphors, and values.

The ontology makes these relationships explicit: the `RELATED_TO` edges connecting concept nodes, the `THEME` properties grouping concepts into cultural domains, the `EXEMPLIFIED_BY` links from concepts to proverbs. When OG-RAG queries "proverbs about wealth," it retrieves not just proverbs containing "wealth" keywords, but proverbs connected to the `Material_Wealth` concept node—which itself connects to related concepts like `Social_Capital`, `Community_Responsibility`, and `Inheritance_Patterns`.

Traditional RAG, given the same query, retrieves based on embedding similarity—a vector space approximation of these relationships, but without explicit structure. The 5.3% performance gap quantifies the value of explicitness: structured knowledge enables more precise retrieval, fewer false positives, and better handling of edge cases where embeddings fail (polysemy, rare terms, dialectical variants).

Yet the Structure Hypothesis faces an immediate objection: if structure is so advantageous, why did Raw GPT-4—operating purely on learned embeddings without explicit structure—achieve 91% of OG-RAG's cultural authenticity? The gap is statistically significant but hardly a chasm. One could argue that modern LLMs have learned sufficient cultural structure *implicitly* through web-scale pretraining, rendering explicit ontologies redundant.

The rebuttal lies in examining failure modes rather than aggregate scores. Raw GPT-4's errors weren't randomly distributed—they systematically concentrated on proverbs with high metaphorical density (18% average error rate vs. 8% for literal proverbs) and domain-specific terminology (22% error rate). When OG-RAG succeeded where GPT-4 failed, the difference traced to precise ontological relationships: the stork-locust-wealth proverb required knowing that financial vigilance metaphors connect to `Economic_Wisdom` through `Animal_Behavior_Analogy` nodes—a three-hop graph relationship encoded explicitly in the ontology but only probabilistically approximated in embedding space. Structure doesn't replace learned representations—it *complements* them by providing precision where embeddings offer only approximation.

**Trade-offs and Engineering Considerations** The Structure Hypothesis comes with caveats. Does the 5.3% performance gain justify the engineering cost? Ontology construction required approximately 200 person-hours of expert effort (cultural consultants + knowledge engineers) to create the 847-concept, 1,200-relationship structure. Traditional RAG required only collecting proverbs ( 20 hours) and generating embeddings (automated). For resource-constrained projects, this 10x labor differential may be prohibitive.

Yet the cost-benefit calculus shifts when we examine error patterns. Our analysis (Section 5.5.2) revealed that 68% of OG-RAG's remaining errors stem from *ontology incompleteness*—concepts missing from the graph. The error analysis points toward an incremental development strategy: start with a small core ontology (say, 100-200 concepts covering the most common cultural themes), deploy the system, collect error cases, expand the ontology based on observed failures, and iterate. Such incremental expansion amortizes engineering effort over time while delivering immediate benefits.

Moreover, the ontology is *reusable* across multiple downstream tasks—not just translation, but cultural education, semantic search, digital heritage applications. Traditional RAG's text corpus is task-specific. The long-term return on ontology investment may exceed the upfront cost when multi-task utility is considered.

**When Structure Helps vs. When It Doesn't** Not all translation tasks benefit equally from ontological structure. The advantage correlates strongly with three proverb characteristics. **Metaphorical density** matters most: compound metaphors (like the stork-locust-wealth proverb) improved 18% with OG-RAG versus just 8% for literal proverbs. **Domain-specific terminology** follows close behind—Kikuyu-specific terms like *ngwataniro* (communal work party) saw 22% gains compared to 7% for common vocabulary. **Implicit cultural references** requiring background knowledge (clan hierarchies, ritual practices) improved 16% versus 5% for self-contained proverbs.

Conversely, simple proverbs with transparent meanings (e.g., *"Ciira muingi ni ukia"*—"Many lawsuits bring poverty") showed minimal difference: OG-RAG scored 0.521, Traditional RAG 0.507. This pattern suggests a hybrid strategy—deploy structured retrieval for complex, culturally dense texts; fall back to simpler methods when complexity doesn't warrant the overhead.

## 6.2 Positioning Within Related Work

### 6.2.1 Cultural NLP and Low-Resource Translation

The thiLLMo system contributes to an emerging research agenda at the intersection of cultural preservation and natural language processing. Recent work has increasingly recognized that mainstream NLP systems—trained predominantly on English and other high-resource languages—fail to capture the cultural nuances of low-resource languages [Nekoto et al., 2020, Joshi et al., 2020].

Our work aligns with and extends several research streams:

**Knowledge-Enhanced Machine Translation** Moussallem et al. [2018] showed that integrating knowledge graphs into neural machine translation improves handling of named entities and domain-specific terminology. However, their focus was factual knowledge (Wikipedia entities, geographic locations) rather than cultural concepts—a crucial distinction. Our cultural ontology represents a different knowledge type: values, metaphors, thematic associations—*cultural semantics* rather than encyclopedic facts.

Zhao et al. [2020] established that injecting BabelNet synsets into transformer encoders improves cross-lingual transfer for 100+ languages. While promising, their approach treats knowledge as disambiguated word senses—still fundamentally lexical. Cultural concepts often transcend individual lexical items; the Kikuyu concept `Uthoni` (reciprocal obligation in kinship networks) has no single-word English equivalent, yet appears across dozens of proverbs.

ThiLLMo's contribution is demonstrating that *cultural knowledge graphs*—with concepts defined by their relational position rather than lexical grounding—can be operationalized in retrieval-augmented generation architectures. This suggests a pathway for extending knowledge-enhanced MT to genuinely cross-cultural scenarios.

**Low-Resource Language Technologies** The African NLP community has pioneered approaches for building language technologies with scarce data [Orife et al., 2020]. The Masakhane initiative, for instance, developed MT systems for 40+ African languages using transfer learning and multilingual models [Abbott et al., 2022].

However, most low-resource MT work focuses on *linguistic* scarcity (limited parallel corpora) rather than *cultural* scarcity (inadequate cultural knowledge representation). Even when African language models achieve decent BLEU scores, they often produce culturally inauthentic translations—replacing indigenous metaphors with Western equivalents, flattening cultural nuance.

Our evaluation framework (Section 5.2) explicitly measured cultural authenticity as distinct from translation fidelity, revealing that Raw GPT-4 achieved 90% of OG-RAG's fidelity but only 91% of its authenticity. This gap would be invisible to BLEU-based evaluation, yet represents precisely the cultural dimension that heritage preservation efforts prioritize.

We advocate for low-resource NLP to expand its evaluation paradigm: linguistic adequacy is necessary but insufficient for cultural text processing. Yet even recent cultural NLP work exhibits a troubling pattern—cultural authenticity is often treated as a *binary property* (culturally appropriate vs. inappropriate) rather than a *gradient phenomenon* with multiple dimensions. A translation might preserve metaphorical imagery while failing value hierarchies, or capture literal meaning while losing pragmatic force. Our multi-dimensional evaluation framework—decomposing quality into authenticity (60%) and fidelity (40%)—emerged from recognizing this complexity. The field needs more nuanced cultural metrics, not just better models. Frameworks must assess cultural preservation along multiple dimensions—requiring partnerships with indigenous knowledge holders, as our expert validation process exemplified.

### 6.2.2 Retrieval-Augmented Generation Architectures

RAG systems have rapidly evolved since Lewis et al. [2020]'s foundational work introducing the paradigm. Recent advances include multi-hop reasoning [Khattab et al., 2021], query decomposition [Press et al., 2022], and hybrid dense-sparse retrieval [Ma et al., 2023].

ThiLLMo's architectural innovation lies in integrating *graph-structured* retrieval into RAG. While previous work explored knowledge graph integration [Sun et al., 2020, Yasunaga et al., 2021a], these efforts focused on question-answering over factoid knowledge (e.g., Ẅho is the CEO of Company X?̈). Cultural translation requires retrieving *conceptual analogues*—proverbs that share thematic essence despite lexical divergence—which factoid KGs are not designed to support.

Our two-stage retrieval process (Cypher query for concept-based proverb retrieval, followed by LLM generation with retrieved context) demonstrates that RAG can be specialized for cultural domains without abandoning the core retrieve-then-generate paradigm. The architectural principle points toward broader applicability: legal RAG systems might retrieve case precedents via legal ontologies; medical RAG might use disease taxonomy graphs.

The 19.8% fidelity improvement over Traditional RAG validates that structured retrieval enhances not just cultural dimensions but semantic precision generally—a finding with relevance beyond cultural applications.

### 6.2.3 Ontology Engineering for Cultural Domains

Cultural ontology development faces unique challenges absent from domain ontologies in medicine [Bodenreider, 2004] or e-commerce [Hepp, 2008]. Cultural concepts often resist formal definition—they are *polythetic* (defined by family resemblance rather than necessary/sufficient conditions) and *context-dependent* (meaning shifts based on social situation).

Our iterative ontology development process (Section 3.4 of the thesis) incorporated three methodological innovations to address cultural ontology's unique challenges. Rather than attempting comprehensive cultural modeling, we extracted concepts *bottom-up* from attested proverbs (**proverb-centric extraction**), ensuring ontological coverage aligned with actual language use rather than idealized cultural categories. We then confronted the issue that cultural concepts often resist crisp boundaries—`Wisdom` and `Elderhood` blend into one another rather than occupying discrete semantic spaces. Our solution: *prototypicality scores* on `EXEMPLIFIED_BY` edges indicating how central a proverb is to each concept (**fuzzy boundary management**). Finally, recognizing that Kikuyu culture is not monolithic, we embedded dialect metadata (Kiambu, Nyeri, Murang'a variants) directly into the ontology (**dialectical variation tracking**), preventing the false precision that comes from modeling cultural knowledge as uniformly interpreted.

These innovations address criticisms of ontological approaches as overly rigid for cultural domains [Almeida et al., 2019]. Cultural ontologies can be *structured yet flexible*—formal enough to enable computational reasoning, loose enough to accommodate cultural complexity.

## 6.3 Limitations and Ethical Considerations

### 6.3.1 Technical Limitations

Despite OG-RAG's demonstrated effectiveness, several technical limitations constrain generalizability and practical deployment:

**Ontology Incompleteness** Our 847-concept ontology covers approximately 60-70% of concepts in the broader Kikuyu proverb corpus (estimated 2,000+ proverbs in published collections). Error analysis traced 68% of remaining failures directly to missing concepts—particularly spiritual practices, specific agricultural tools, and inter-ethnic relations.

Ontology incompleteness is an inherent challenge: cultural knowledge is open-ended and constantly evolving. What surprised us most during error analysis was discovering that a handful of missing concepts—perhaps 15-20 high-frequency cultural frames like `Intergenerational_Knowledge_Transfer` and `Ritual_Timing`—accounted for the majority of failures. This concentration suggests targeted ontology expansion could yield disproportionate improvements. Proposed mitigation strategies include: (1) active learning approaches that identify high-impact missing concepts based on translation errors, (2) community-driven ontology expansion tools allowing cultural practitioners to contribute knowledge, and (3) hybrid architectures that gracefully degrade to Traditional RAG when encountering unknown concepts.

**Dialectical and Generational Variation** The evaluation dataset combined proverbs from three Kikuyu dialects, with expert translations representing a ”standard” Kikuyu interpretation. However, 5% of proverbs showed inter-annotator disagreement reflecting genuine dialectical differences. Additionally, younger Kikuyu speakers (¡ 40 years old) sometimes interpret proverbs differently from elders due to changing cultural contexts (e.g., proverbs about cattle wealth reinterpreted for urban settings).

The current system does not model this variation systematically. Future work should explore *multi-perspective* ontologies that represent competing cultural interpretations rather than seeking a single ”correct” representation.

**Scalability Beyond Proverbs** While the architecture is theoretically applicable to other cultural text genres (folktales, songs, ceremonial discourse), proverbs have unique properties—brevity, metaphorical density, decontextualizability—that may not generalize. Folktales, for instance, require modeling narrative structure and character archetypes, not just cultural concepts. Ceremonial discourse involves performative context (ritual timing, social roles) absent from written proverbs.

Each genre likely requires genre-specific ontological modeling. The cultural concept ontology developed here provides a *foundation* but not a complete solution for comprehensive cultural text processing.

### 6.3.2 Ethical Considerations

Developing AI systems for cultural heritage involves profound ethical responsibilities:

**Epistemic Authority and Knowledge Ownership** Who has the authority to define "authentic" Kikuyu cultural knowledge? Our ontology was developed with Kikuyu cultural experts, but their perspectives—while deeply informed—represent particular positionalities (educated, internationally-engaged individuals). Grassroots community members, rural elders, and diaspora populations might contest certain ontological modeling choices. The hardest choice we faced was deciding when cultural disagreement reflected genuine dialectical variation (which should be preserved in the ontology as competing interpretations) versus individual idiosyncrasy (which might mislead the system).

The risk of *epistemic freezing* looms: once cultural knowledge is formalized in an ontology and deployed in AI systems, that representation may acquire undue authority, marginalizing alternative interpretations. We address this through transparent versioning (ontology version 1.0.0, experts consulted documented in metadata) and explicitly framing the system as a *tool for cultural preservation*, not an authoritative cultural arbiter.

Furthermore, all ontology development involved informed consent protocols, compensation for expert time, and co-authorship opportunities on resulting publications. Cultural knowledge is not "raw data" to be extracted—it is intellectual property deserving appropriate recognition.

**Cultural Appropriation and Benefit Sharing** AI systems that process indigenous cultural knowledge risk enabling cultural appropriation—non-Kikuyu actors using the system to commodify Kikuyu cultural products without community benefit. While the system itself is open-source (code available on GitHub), we released the cultural ontology under a restricted license requiring non-commercial use without explicit community permission, attribution to Kikuyu cultural knowledge holders, and benefit-sharing agreements for commercial applications (royalties to Kikuyu cultural organizations). These restrictions balance open research ideals with protection of indigenous intellectual property—a tension ongoing in digital heritage ethics [Christen, 2012].

**Translation as Cultural Mediation** All translation involves interpretation; machine translation is no exception. By automating Kikuyu-to-English proverb translation, we potentially shape how non-Kikuyu audiences perceive Kikuyu culture. Subtle translation choices—e.g., rendering *indo* as "wealth" versus "possessions" versus "blessings"—carry cultural implications.

The system includes confidence scores and alternative translations for each output, encouraging users to treat translations as *suggestions* rather than definitive renderings. Additionally, the web interface (future work) will display original Kikuyu text alongside translations, enabling bilingual users to critique translation quality.

Ultimately, no AI system can substitute for human cultural brokers—bilingual individuals who navigate cultural boundaries with contextual sensitivity. ThiLLMo is designed to *assist*, not *replace*, such human mediation.

### 6.3.3 Generalizability to Other Cultural Contexts

While developed for Kikuyu proverbs, the methodology—ontology-grounded RAG for culturally faithful translation—invites extension to other indigenous and low-resource languages. Several factors affect transferability:

**Favorable Conditions for Transfer** The methodology transfers most readily to languages with rich proverbial traditions (East African, West African, Southeast Asian cultures), existing ethnographic documentation enabling ontology bootstrapping, active speaker communities for expert validation, and cultural concepts organized thematically around domains like agriculture, kinship, and governance. These conditions characterize many cultural contexts, suggesting broad applicability.

**Challenges for Transfer** The approach faces steeper challenges with extreme low-resource languages (¡ 1,000 documented proverbs), predominantly oral traditions offering limited textual sources, cultural knowledge held by very few elders (expert scarcity), and culturally sensitive knowledge subject to sacred or restricted access protocols. Such contexts require methodological adaptation: smaller ontologies (200-300 concepts), oral collection protocols, and careful navigation of cultural protocols around knowledge sharing.

The Maasai, Luo, and Luhya languages (also Kenyan) represent natural next targets, given cultural proximity to Kikuyu and availability of documentation. Broader African expansion—Yoruba, Igbo, Zulu—would test cross-cultural transferability of the architectural approach.

### 6.3.4 Theoretical Implications for Cultural AI

Beyond immediate technical contributions, this work surfaces broader questions for the emerging field of *Cultural AI*—artificial intelligence systems designed to engage with human cultural diversity:

**The Limits of Data-Driven Cultural Learning** Contemporary large language models learn cultural knowledge implicitly through web-scale pretraining. Raw GPT-4's competitive performance (91% of OG-RAG's authenticity score) demonstrates that LLMs capture substantial cultural information. Yet the remaining 9% gap—statistically significant and consistent—suggests *limits to implicit cultural learning.*

Cultural concepts are not merely statistical patterns in text; they involve shared understandings, value commitments, and interpretive frameworks that may not be fully recoverable from distributional semantics. The ontology represents cultural knowledge *as structured relationships*—a form of knowledge potentially orthogonal to what LLMs learn.

This suggests a research direction: *hybrid cultural AI* combining learned representations (LLM embeddings) with symbolic structures (cultural ontologies). Each compensates for the other's weaknesses—symbols provide precision and interpretability; neural nets provide coverage and flexibility.

**Cultural Preservation vs. Cultural Dynamism** Heritage preservation efforts often emphasize *conservation*—capturing cultural knowledge as it exists. But cultures are living, evolving systems. Contemporary Kikuyu speakers creatively adapt traditional proverbs, coin new ones, and reinterpret old ones for modern contexts.

Should a cultural AI system represent culture as *stable tradition* (preservationist stance) or *dynamic process* (vitalist stance)? Our versioned ontology (allowing temporal evolution tracking) gestures toward the latter, but the tension remains unresolved. How can AI systems support—rather than ossify—cultural dynamism? The question points toward future architectures that model not just cultural knowledge but *cultural change processes.*

**From Representation to Engagement** Most cultural NLP work treats culture as something to be *represented* (in ontologies, in training data). But from anthropological perspectives, culture is enacted through *practice*—dialogue, ritual, storytelling [Bourdieu, 1977].

Future cultural AI might move from representation to *engagement*: systems that participate in cultural practices (interactive storytelling, language learning dialogues, ceremonial assistance) rather than merely processing cultural texts. This requires rethinking AI architecture—from static knowledge bases to dynamic interactional systems.

The discussion has contextualized thiLLMo's contributions within theoretical frameworks, related work, and ethical considerations. Chapter 7 synthesizes key findings, articulates concrete contributions, and proposes directions for future research in ontology-guided cultural AI systems.

# Chapter 7

# Conclusion and Future Work

Can cultural ontologies enhance retrieval-augmented generation for low-resource translation? This thesis answered affirmatively for Kikuyu proverb translation, demonstrating that ontology-grounded RAG (OG-RAG) achieves statistically significant improvements: 10.5% in cultural authenticity, 19.8% in translation fidelity, 13.5% in overall quality. But the more interesting question isn't whether it works—it's why structure matters, what this reveals about cultural knowledge representation, and where these insights might lead. This chapter synthesizes contributions (Section 7.1), acknowledges limitations honestly (Section 7.2), and charts future directions (Section 7.3) that extend beyond technical refinement toward genuine cultural impact.

## 7.1 Research Contributions

This work makes contributions across technical, methodological, and theoretical dimensions:

### 7.1.1 Technical Contributions

**Ontology-Grounded RAG Architecture** The primary technical contribution is the thiLLMo system architecture, which integrates a domain-specific cultural ontology with retrieval-augmented generation for machine translation. The architecture introduces **graph-based retrieval** replacing vector similarity search with Cypher queries over

Neo4j knowledge graphs—enabling retrieval based on cultural concept relationships rather than lexical similarity. This foundational shift enables **concept-proverb linking** via `EXEMPLIFIES` relationships, allowing thematic retrieval independent of surface form. Recognizing that no ontology achieves complete coverage, we implemented a **hybrid fallback strategy** that gracefully degrades to Traditional RAG when encountering unknown concepts, maintaining robustness while maximizing structured knowledge utilization. Finally, **multi-dimensional prompt engineering** explicitly incorporates retrieved cultural context, ontological concept descriptions, and example translations to guide culturally faithful output.

The architecture is modular and language-agnostic, facilitating adaptation to other low-resource cultural translation scenarios. Open-source release (GitHub: `ndethi/opit-rai9001`) enables reproducibility and community extension.

**Kikuyu Cultural Ontology** The development of a 847-concept, 1,200-relationship cultural ontology for Kikuyu proverbs represents a substantial digital heritage artifact. The ontology:

- Covers 15 thematic domains (Agriculture, Kinship, Governance, Wisdom, Economics, etc.)
- Represents cultural concepts via graph relationships (`RELATED_TO`, `EXEMPLIFIES`, `BROADER_THAN`)
- Incorporates dialectical variation metadata (Kiambu, Nyeri, Murang'a dialects)
- Includes 100+ high-quality expert-validated Kikuyu proverbs with English translations
- Provides a template for ontology engineering in other cultural domains

Beyond immediate translation applications, this ontology serves ongoing digital humanities projects, Kikuyu language education initiatives, and cultural preservation efforts by the Kikuyu community.

### 7.1.2 Methodological Contributions

**Cultural Translation Evaluation Framework** Standard machine translation metrics (BLEU, CHRF++, METEOR) prioritize lexical overlap and surface similarity, often failing to capture cultural authenticity—the dimension most critical for heritage preservation. Our evaluation framework introduced:

- **Decomposed Quality Metrics**: Separating cultural authenticity (preservation of cultural frames, metaphors, values) from translation fidelity (semantic accuracy), recognizing these as orthogonal desiderata.
- **Expert-Validated Gold Standards**: 100 proverbs with human expert translations serving as evaluation benchmarks, developed through rigorous annotation protocols with 94% inter-annotator agreement.
- **Weighted Composite Scoring**: 60% authenticity / 40% fidelity weighting derived from cultural expert consultations, reflecting community priorities for cultural preservation over literal accuracy.
- **Statistical Rigor**: Paired t-tests, effect size calculations, and distribution analysis providing robust statistical validation beyond point estimates.

This framework is transferable to evaluating other cultural text processing systems (poetry translation, ceremonial discourse analysis, folktale generation) and addresses a significant methodological gap in cultural NLP research.

**Iterative Ontology Development Process** Cultural ontology engineering lacks established best practices, with most work adapting methodologies from biomedical or enterprise domains. Our bottom-up, proverb-centric development process contributes several innovations:

- **LLM-Assisted Concept Extraction**: Using GPT-4 to propose candidate concepts from proverb collections, with human expert validation—reducing manual effort by 40% while maintaining quality.

- **Prototypicality-Based Fuzzy Boundaries**: Representing cultural concepts via exemplar proverbs with graded typicality scores rather than formal definitions, accommodating polythetic categories.

- **Versioned Community Co-Creation**: Transparent version control (Git-based), documented provenance of expert contributions, and pathways for community expansion—addressing epistemic authority concerns.

These methodological innovations make cultural ontology development more tractable for resource-constrained heritage projects.

### 7.1.3 Theoretical Contributions

**The Structure Hypothesis for Cultural Knowledge** Chapter 6 articulated the **Structure Hypothesis**: for culturally grounded text generation, structured knowledge representations (ontologies) outperform unstructured representations (text corpora, embeddings) when cultural concepts exhibit hierarchical, relational, or thematic organization. The 5.3% performance gap between OG-RAG and Traditional RAG—identical architectures differing only in retrieval index structure—provides empirical support.

This hypothesis has implications for broader AI research: neural embedding spaces, despite their power, may be *necessary but insufficient* for cultural reasoning. Hybrid approaches combining learned representations (neural nets) with symbolic structures (knowledge graphs) warrant further investigation.

**Cultural AI Design Principles** The ethical analysis (Section 6.3.2) surfaced design principles for *Cultural AI*—systems engaging with indigenous and minority cultural knowledge. These principles emerged from concrete challenges we faced during development, not abstract ethical theorizing. **Community partnership** became non-negotiable once we recognized that cultural knowledge is intellectual property requiring informed consent, fair compensation, and co-authorship—not "raw data" for extraction. **Epistemic humility** followed from observing disagreements among cultural experts: AI systems must present cultural knowledge as *perspectives* (versioned, attributed, contestable) rather than authoritative facts. **Benefit sharing** addresses the uncomfortable reality that AI systems can commodify cultural knowledge; licensing and governance structures must ensure communities benefit. Finally, **graceful degradation** reflects the observation that cultures are internally diverse—systems must handle variation robustly rather than forcing monolithic representations.

These principles inform responsible development of cultural technologies, an increasingly urgent concern as AI expands into heritage domains.

## 7.2 Limitations Revisited

Every research design makes trade-offs; acknowledging these honestly contextualizes contributions and guides future work.

### 7.2.1 Scope and Generalizability

The research deliberately narrowed scope to one language (Kikuyu), one genre (proverbs), and one translation direction (Kikuyu→English). This focus enabled depth, but at the cost of breadth. While the methodology is designed for transferability, empirical validation in other cultural contexts remains future work—and may reveal that some architectural choices were Kikuyu-specific rather than generalizable. Proverbs' unique properties—brevity, metaphorical density, decontextualizability—may not extend to longer or less formulaic cultural texts like folktales or ceremonial discourse.

### 7.2.2 Ontology Coverage

The 847-concept ontology covers an estimated 60-70% of concepts in the broader Kikuyu proverb corpus. The remaining 30-40% includes specialized domains (spiritual practices, specific clan customs, rare metaphors) that require further expert consultation to model. Error analysis attributed 68% of system failures to this incompleteness.

Ontology development is inherently iterative; version 1.0.0 should be viewed as a foundation requiring ongoing community-driven expansion rather than a complete cultural model.

### 7.2.3 Evaluation Scale

The 100-proverb evaluation dataset, while sufficient for statistical significance, represents ¡5% of documented Kikuyu proverbs ( 2,000+ in published collections). Broader evaluation would increase confidence in generalizability and might reveal edge cases not captured in the current sample.

Resource constraints (expert translator time, annotation costs) limited evaluation scale. Future work with institutional support could expand to 500+ proverbs across more diverse sources (oral collections, regional dialects, historical texts).

### 7.2.4 Dialectical and Generational Variation

The current system models a ”standard” Kikuyu interpretation derived from three dialects (Kiambu, Nyeri, Murang'a) and primarily elder expert perspectives. Significant intra-cultural variation exists—younger speakers interpret proverbs differently, urban vs. rural contexts shape meanings, and diasporic communities adapt traditions.

Multi-perspective cultural AI systems that represent *diversity within cultures*, not just *diversity across cultures*, remain an open challenge.

## 7.3 Future Research Directions

Several promising avenues extend this work:

### 7.3.1 Cross-Cultural Validation and Transfer

**Immediate Extensions: Related Kenyan Languages** Applying the methodology to closely related languages (Kikamba, Kimeru, Kiembu—all Central Bantu) would test architectural transferability while controlling for cultural similarity. These languages share many cultural concepts with Kikuyu, potentially enabling *ontology transfer* with incremental adaptation rather than building from scratch.

**Broader African Language Exploration** Extending to more distant African languages—Yoruba (Niger-Congo), Swahili (Bantu, but coastal/Islamic influences), Amharic (Afroasiatic)—would test cross-cultural robustness. Each has rich proverbial traditions and existing linguistic documentation, but different cultural organizations requiring new ontological modeling approaches.

**Beyond Africa: Global Indigenous Languages** Languages like Maori, Quechua, Navajo, and Hawaiian face similar cultural preservation challenges. Collaborations with indigenous communities in Oceania, Americas, and Asia could establish whether ontology-grounded RAG represents a *general solution* for low-resource cultural translation or remains Africa-specific.

### 7.3.2 Genre Expansion

**Folktales and Oral Narratives** Kikuyu folktales involve longer texts, narrative structure, character archetypes, and moral frameworks. Extending OG-RAG to folktale translation would require ontological modeling of narrative patterns—perhaps via story grammars [Rumelhart, 1975] integrated with cultural concept graphs.

**Ceremonial and Ritual Discourse** Wedding speeches, initiation rites, and funeral orations employ highly formulaic language with strict performative contexts. Such texts require modeling *pragmatic context* (social roles, ritual timing, performative force) beyond semantic content—a significant architectural extension.

**Contemporary Cultural Production** Modern Kikuyu cultural expression—hip-hop lyrics, social media proverbs, political discourse—creatively adapts traditional forms. Modeling *cultural innovation* requires dynamic ontologies that track concept evolution over time, perhaps via temporal knowledge graphs [Leblay and Chekol, 2018].

### 7.3.3 Technical Enhancements

**Active Learning for Ontology Expansion** Rather than manual ontology construction, *active learning* approaches could identify high-impact missing concepts: deploy the system, collect translation failures, cluster errors to identify recurring gaps, query experts about high-frequency gaps, iteratively expand ontology. This could reduce human effort by 50-70% while maintaining coverage.

**Multi-Modal Cultural Knowledge** Cultural knowledge is not purely textual—it includes visual symbols, musical patterns, material artifacts, and embodied practices. *Multi-modal ontologies* linking text concepts to images (e.g., traditional Kikuyu architecture), sounds (e.g., ceremonial music), and gestures (e.g., greeting protocols) could support richer cultural AI applications.

Vision-language models (CLIP, GPT-4V) provide technical foundations; the research challenge is cultural modeling: How do visual and textual cultural knowledge interact? Can multi-modal retrieval improve translation of proverbs referencing material culture?

**Interpretable Retrieval Explanations** Currently, the system retrieves relevant proverbs but does not *explain* retrieval decisions to users. Adding *interpretability*: highlighting which ontological paths triggered retrieval, displaying concept relationships, showing alternative retrieval options—would increase user trust and enable cultural learning.

Graph visualization libraries (Cytoscape.js, Neo4j Bloom) provide UI components; the challenge is cognitive: What explanations are most useful for non-expert users engaging with cultural AI?

### 7.3.4 Practical Deployment and Impact

**Educational Applications** The system could support Kikuyu language education: interactive proverb learning, cultural context explanations, translation practice with feedback. Deployment in Kenyan schools (where Kikuyu is taught as a "mother tongue" subject) could reach thousands of students.

Partnerships with educational NGOs (e.g., African Storybook Initiative) and government education ministries could pilot classroom integration, with effectiveness measured via language proficiency and cultural competency assessments.

**Digital Heritage Platforms** Integration with digital heritage initiatives—online proverb databases, cultural museums, community archives—could democratize access to Kikuyu cultural knowledge. A web platform allowing users to:

- Search proverbs by theme, keyword, or cultural concept
- View translations with cultural context explanations
- Contribute new proverbs or suggest ontology expansions
- Discuss interpretations in community forums

would transform the system from research artifact to living cultural resource.

**Community-Driven Governance** Long-term sustainability requires transferring system governance to Kikuyu cultural organizations. Models like the Maori Language Commission's stewardship of Māori language technologies [Keegan et al., 2017] provide templates: community boards overseeing ontology updates, cultural licensing, and benefit sharing from commercial applications.

Technical implementation (cloud hosting, maintenance, feature development) could be provided by research teams, but *cultural authority*—decisions about what knowledge to include, how to represent it, who benefits—must rest with communities.

### 7.3.5 Theoretical Research Questions

**Limits of Ontological Formalization** How much cultural knowledge *can* be formalized in ontologies? Some knowledge is tacit, performative, or context-dependent in ways resisting symbolic representation. Ethnographic research investigating what cultural practitioners consider "(un)formalizable" could inform realistic expectations for cultural AI.

**Cultural Ontology Dynamics** If cultures constantly evolve, how should ontologies represent change? Approaches might include: temporal versioning (ontology snapshots over time), probabilistic concept definitions (concepts with changing prototypes), or dialectical contradictions (explicitly representing contested concepts). Research questions: Which approach best supports cultural vitality? How do users engage with evolving vs. static cultural knowledge?

**Cross-Cultural Ontology Alignment** Can ontologies from different cultures be *aligned*—finding correspondences between Kikuyu concepts and, say, Yoruba or Maori concepts? Successful alignment could enable cross-cultural comparison, translation between indigenous languages (bypassing English), and identification of universal vs. culture-specific patterns.

Techniques from ontology matching [Euzenat and Shvaiko, 2007] provide starting points, but cultural alignment poses unique challenges: concepts may partially overlap, relate analogically rather than equivalently, or be genuinely incommensurable.

## 7.4 Closing Reflections

This research began with a practical question—how can AI systems translate Kikuyu proverbs while preserving cultural authenticity?—and evolved into a broader investigation of how cultural knowledge should be represented, validated, and deployed in computational systems.

The success of ontology-grounded RAG demonstrates that *structure matters*: making cultural relationships explicit improves machine translation quality measurably. But beyond technical validation, the research surfaced deeper questions: Whose cultural knowledge gets formalized? Who benefits from cultural AI systems? How can technology support cultural vitality rather than ossification?

These questions have no purely technical answers. They require ongoing dialogue between computer scientists, cultural practitioners, and communities whose heritage is at stake. This thesis contributes one model for such collaboration—methodologically rigorous yet ethically reflexive, technically innovative yet culturally grounded. But it's just one model, and likely not the best for all contexts.

As AI systems increasingly engage with human cultural diversity—through translation, content generation, digital heritage, and beyond—the need for *culturally-aware architectures* will intensify. Ontology-grounded approaches represent one promising pathway, proven effective for structured cultural domains. But many others await exploration: participatory design methodologies where communities co-create systems from inception, multi-perspective knowledge representation that embraces internal cultural diversity rather than seeking consensus, community-governed AI systems where cultural authority rests with heritage communities rather than technical experts. The field is young enough that radically different approaches remain viable.

The ultimate measure of success is not technical performance alone, but *cultural impact*: Does this work support Kikuyu language vitality? Does it empower cultural practitioners? Does it foster cross-cultural understanding while respecting cultural specificity?

Preliminary engagement with Kikuyu educators and cultural organizations suggests this work might transcend academic contribution. The ontology has been requested for integration into a Kikuyu language learning mobile app reaching 5,000+ students; the proverb corpus has been cited in a community cultural documentation project; two Kikuyu cultural consultants have expressed interest in co-authoring papers on the methodology. One consultant noted: "This isn't just about technology—it's about our children knowing where they come from."

These early signs suggest that technically rigorous cultural AI can also be *socially valuable*—a convergence that should guide future work in this emerging field. The real test isn't whether the system achieves statistical significance, but whether it serves cultural vitality.

The tools exist. The questions are clear. The communities aren't just waiting—they're already building.

The next chapter—beyond this thesis—belongs to collaborative cultural innovation. This work offers one foundation; what gets built on it depends on partnerships yet to form, questions yet to ask, and communities whose voices will shape what cultural AI becomes.

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